

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

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ABSTRACT. We formulate a common foundation for controlled statistical experiments and attainable prediction without assuming a finite-dimensional state, a Fisher metric, or a large-deviation principle. Future tests define a minimal predictive quotient whenever the required measurable realization exists. Experiment reductions are implemented by parameter-independent causal machines; their errors compose in total variation and their internal states are charged to the memory budget. We prove a finite-state emulation theorem that combines state quantization, numerical perturbation, and feedback-uniform path-law comparison, including stopped experiments. A separate quantization identity characterizes checkpoint squared-prediction regret under the actual preparation law. For a refreshing hidden Markov chain with continuously commanded binary observations, we derive matching bounds of order $\epsilon^2 M^{-2/d}$ for every integer memory budget M , uniformly over all positive times and through vanishing observation strength $0 \leq \epsilon \leq 1/4$. One finite-state filter attains the upper bound; the lower bound follows from the exact Jacobian of the physical posterior map, with failure reports retained. If the terminal query is attenuated as well, the order becomes $\epsilon^4 M^{-2/d}$. We also establish the appropriate posterior, relative-entropy, score, and exponential-transform identities, and give explicit examples separating sufficiency, information, causal memory, and exponential asymptotics. The abstract identities are attributed to their classical antecedents; the quantitative results have explicit model, preparation, and resource assumptions.

CONTENTS

1. Introduction	3
1.1. The quantitative results	3
1.2. Objects and reductions	4
1.3. Relation to established theories	5
1.4. Reading the paper	5
2. Positive causal experiments	5
2.1. Types and primitive data	5
2.2. Strategies, paths, and stopping	6
2.3. Prepared and parameter-uniform questions	7
2.4. Instruments and evidence	7
2.5. Preparing a conditional experiment	7
3. Future tests and predictive quotients	8
3.1. Operator words and adaptive protocols	8
3.2. Safe measurable realizations	9
3.3. Finite remaining-test coordinates	9
3.4. Three different fibres	10

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4. Executable reductions and charged resources	10
4.1. Finite-state implementations	10
4.2. The path-kernel estimate	10
4.3. Checking an exact state reduction locally	12
4.4. Terminal and causal equivalence	12
5. Posterior geometry and information loss	12
5.1. Posterior barycentres	12
5.2. Relative entropy and recovery	13
5.3. Scores and Fisher information	13
5.4. Sequential information in finite positive models	14
6. Attained prediction and checkpoint quantization	14
6.1. The physical loss	14
6.2. A small-ball lower bound	15
6.3. Whole-image covers and all integer budgets	16
6.4. Causal lower bounds use one exploration law	16
6.5. Calibrated anisotropy	16
7. Causal quantization with feedback and stopping	16
7.1. Admissible state domains	16
7.2. What feedback does and does not require	18
7.3. Uniform prediction and finite-horizon path comparison	18
7.4. Model or calibration perturbations	18
8. A dynamic model with a uniform memory law	18
8.1. The experiment and its resource convention	18
8.2. Explicit posterior recursion	20
8.3. The actual acquisition density	21
8.4. A fixed codebook and the proof of the theorem	22
8.5. Feedback-uniform emulation in this model	22
8.6. Calibration and finite precision	23
9. Normalized response and observation modules	23
9.1. The covariance pairing	23
9.2. Finite positive functionals and exact product fibres	24
9.3. The contact calculation	24
9.4. Why general dynamic tests need not form a finite algebra	25
10. Exponential transforms and conditional limit interfaces	25
10.1. Random channels have nonlinear exponential pullbacks	25
10.2. A finite risk-sensitive recursion	26
10.3. Deterministic LDP contractions	27
10.4. Regular quadratic and labelled-phase interfaces	27
10.5. Closure and finite-dimensional memory	27
11. Boundary examples and non-implications	28
12. Interfaces with singular geometry and dynamics	29
12.1. What is established in this installment	29
12.2. The role of the two geometric manuscripts	30
12.3. A precise next target	30
12.4. The physical realization obligations remain	30
12.5. Conclusion	31
Appendix A. Measurability, finite covers, and a scalar benchmark	31
A.1. Finite-label implementation	31
A.2. Conditional distributions and common versions	31

A.3. Quantization after a readout change	31
A.4. Predictable error budgets and random horizons	32
Appendix B. Explicit finite checks and the limits of computation	32
B.1. A two-state posterior calculation	32
B.2. A direct resource calculation	33
B.3. A finite precision requirement rather than a free real tape	33
B.4. Role of the accompanying diagnostics	33
References	34

1. INTRODUCTION

An observer does not retain a physical state merely by naming it. The observer receives a history, chooses actions, and stores a representation from which future decisions or predictions must be made. There are consequently three different questions. Which future laws are determined by that history? Which parts of those laws can a second experiment reproduce? How much persistent state is required to reproduce them causally? The answers need not be ordered by a single notion of information.

This paper starts with a positive causal experiment and separates these three questions. The starting data are a preparation, transition–observation kernels, executable future tests, admissible interventions, and a resource interface. A posterior, a predictive quotient, and the law of attained predictions are derived from these data. Smooth structures, information matrices, asymptotic rates, and operator realizations are additional structures, not axioms of the experiment.

The distinction is already visible in a binary example. If a hidden bit is observed exactly, two persistent labels retain its posterior prediction without loss. Adding independent Gaussian noise makes the observation less informative in the statistical order, but the posterior then ranges continuously between zero and one. No fixed finite number of labels preserves all of those posterior predictions exactly. There is no violation of data processing: the two excess losses use different full-observation oracles. The complete calculation appears in Section 11.

1.1. The quantitative results. Our first quantitative result concerns a prepared predictive-state model. Let s_t be its exact state and let

$$s_{t+1} = F_t(s_t, a_t, y_{t+1}).$$

Suppose the same-input update is L_t -Lipschitz on a common admissible domain. Cover that domain by M_t representatives with radius r_t , and allow an implementation error η_t in each update. One causal label machine then has error bounded by the deterministic recurrence

$$b_0 = 0, \quad b_{t+1} = L_t b_t + r_{t+1} + \eta_{t+1}. \quad (1.1)$$

The same machine also defines a finite-state model, provided the next-report law is Lipschitz in the state. If its total-variation Lipschitz constant is κ_t , the entire visible-path laws differ by at most

$$\min \left\{ 1, \sum_{t=0}^{N-1} \kappa_t b_t \right\}, \quad (1.2)$$

uniformly over the common feedback policy class. The policies need not be continuous. They agree on equal observed histories, which is enough for the kernel telescoping proof. For stopped experiments, the sum is charged only while the reference experiment remains active. A policy that itself retains D_t states yields a controller with at most $D_t M_t$ states, not M_t . This is Theorem 7.1. It distinguishes same-input prediction accuracy from closed-loop comparison instead of inferring the latter from the former.

The second result supplies the geometry and the actual acquisition law in a dynamic model rather than postulating them. Fix $d \geq 1$ and let the hidden state take values in $\{0, \dots, d\}$. Between observations it stays put with probability $1 - \gamma$ and is replaced by an independent uniform state with probability γ , where $4/5 \leq \gamma \leq 1$. A command $u \in [-1, 1]^d$ produces a binary report $y \in \{-1, 1\}$ with

$$\mathbb{P}(y \mid X = i, u) = \begin{cases} 1/2, & i = 0, \\ (1 + y\epsilon \tanh u_i)/2, & 1 \leq i \leq d, \end{cases} \quad 0 \leq \epsilon \leq 1/4. \quad (1.3)$$

At a selected checkpoint, a terminal query is drawn only after the label has been formed. Query i returns one binary outcome of mean $1/2 + \tau \mathbf{1}_{\{X=i\}}$, where $\tau = 1/4$ and $1 \leq i \leq d$. All acquisition reports, both signs, are retained in the preparation law. Only one selected terminal query is executed in a run.

Let $R_{M,t}^{\text{chk}}$ be the optimal checkpoint excess squared loss under independent uniform commands, and let $\mathfrak{R}_{M,N}$ be the best maximum-checkpoint excess attainable by one causal machine with at most M labels after each acquisition. Theorem 8.1 proves

$$c_d \epsilon^2 M^{-2/d} \leq R_{M,t}^{\text{chk}} \leq \mathfrak{R}_{M,N} \leq C_d \epsilon^2 M^{-2/d}, \quad 1 \leq t \leq N. \quad (1.4)$$

The middle inequality is understood for the optimal maximum risk, so it holds for each fixed checkpoint. The same upper-bound machine works at every time, and its pathwise prediction error is bounded uniformly over all command and report sequences. Thus the maximum can be replaced by a supremum over all positive times. The constants depend on d and the fixed terminal probe, not on N, t, M, ϵ , or γ in the displayed ranges. At $\epsilon = 0$, one label is exact.

The proof is elementary but combines obligations that should not be suppressed. Refreshing gives an interior lower bound for the pre-observation belief. Bayes updating is uniformly contractive after refreshing. Its entire reachable set lies within distance $O(\epsilon)$ of the uniform belief. The command-to-posterior Jacobian has determinant of order ϵ^d , uniformly over every previous history. It follows that the actual posterior law has a density bounded above by $C_d \epsilon^{-d}$. This gives a lower bound against all M -label encoders, including independent public or private coding randomization. A finite cover of an invariant belief domain gives the matching causal construction. No stationary law of the filter and no asymptotic quantization formula is assumed.

This model has a genuinely evolving latent state. It is not the static latent-variable, conditionally independent acquisition problem of [15]. Its nonpolynomial command coordinate is convenient, not an intrinsic novelty claim: the change $v_i = \tanh u_i$ makes the likelihood affine in v . The additional conclusion relative to that manuscript's stated finite-horizon scope is the time-uniform bound obtained here from explicit refreshing contraction. If the terminal query is also multiplied by ϵ , its physical metric changes and the order becomes $\epsilon^4 M^{-2/d}$ (Corollary 8.5).

1.2. Objects and reductions. A statistical comparison acts first on a family of probability laws. We use $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ to mean that the source experiment $\mathfrak{E}$ can implement $\mathfrak{F}$ by a parameter-independent causal machine. A reduction includes its strategy lift, state, observation interface, and resource certificate. Serial composition adds total-variation defects and composes budget maps. For a common bounded decision loss, this gives a risk inequality. It does not assert that two different posterior oracles have ordered coding losses.

The minimal predictive quotient is defined by equality of all designated future tests, with equal continuation interfaces. It is an exact semantic object. In a finite-state realization its updates are explicit; for a general measurable quotient, standard Borel realizability is a separate obligation. We give a compact continuous realization theorem rather than assuming that every quotient is a smooth state space.

1.3. Relation to established theories. Kernel composition, disintegration, sufficiency, and their categorical organization are established foundations; see Fritz [2] and Fritz–Gonda–Perrone–Rischel [3]. Blackwell’s comparison of experiments [1] concerns decision performance under garbling. Our directional simulation inequalities use that classical idea; we retain the timing and state required to implement a garbling. The introduction of a new project name does not make these results new.

Predictive-state and causal-state representations already define state by future distributions, including action-conditional versions [5, 6, 7]. Our quotient theorem specializes to that principle. The present presentation makes the future test class, preparation, zero-evidence convention, and resource interface explicit, and supplies sufficient conditions for a measurable realization.

Quantized nonlinear filtering and finite model approximation have an extensive literature; examples include [8, 9, 10]. Approximating a signal on a finite grid while storing an arbitrary vector of posterior weights is not the same resource as storing one of M complete prediction labels. Conversely, the cardinality bounds proved here are not bounds on all numerical workspace or on the size of a read-only program. The state error recurrence and the path-kernel comparison use standard stability and telescoping arguments. The contribution of Theorem 7.1 is their explicit joint statement under feedback, stopping, numerical defects, and simulator-state accounting. No assertion is made that these abstract ingredients have no antecedents.

The preservation of Fisher information is local and need not imply sufficiency, as explained by Pollard [11]. We prove the finite-model score formula and state the exact QMD interface. Exponential integral–entropy duality is classical; its proof is included to fix the risk parameter and the nonlinear pullback under a random channel. The contact-fibre calculation of [16] is connected through linear observations of positive functionals, not identified with an entire statistical experiment.

1.4. Reading the paper. Sections 2–4 establish the objects, quotients, and executable reductions. Section 5 fixes the Bayesian, entropic, and score identities. Sections 6–8 contain the quantitative results and their full proofs. Sections 9 and 10 connect normalized response and exponential transforms. Section 11 gives explicit limitations, and Section 12 states the interfaces with subsequent geometry and dynamical work. The appendices contain measurable and finite-experiment details used in the proofs. There is no dependence on an unproved theorem of a later installment.

2. POSITIVE CAUSAL EXPERIMENTS

2.1. Types and primitive data. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. All primitive spaces below are standard Borel. Time is initially $t = 0, \dots, N$, with the countable-time extension stated separately. The unknown parameter is $\lambda \in \Lambda$; it is not a random variable until a prior is supplied. The hidden state, applied action, and next report have spaces X_t, A_t, Y_{t+1} . A visible history is

$$h_t = (a_0, y_1, \dots, a_{t-1}, y_t) \in H_t.$$

A public initial report, clock increments, and observed cost records can be included in this notation. An action-availability rule, when present, is part of the interface. We usually write a common action space to avoid encoding that rule repeatedly.

Definition 2.1 (Raw experiment). A raw experiment consists of a parameter space Λ , preparations $\mu_\lambda \in \mathcal{P}(X_0)$, and jointly measurable probability kernels

$$K_t^\lambda(dx', dy, dc \mid x, a) \tag{2.1}$$

on $X_{t+1} \times Y_{t+1} \times \mathcal{C}$, together with a class of executable future tests and a resource interface. The nonnegative vector c records charged increments, such as raw preparations or physical time. When it is observed, it is included in y ; when it is not observed, it is integrated in the visible law but is

still counted in the resource account. Every kernel has total mass one. A failed measurement and a terminating measurement are reports, not missing mass.

The following six requirements will be referred to as the foundational axioms. They specify the admissible language rather than analytic conclusions.

- A0 (measurable interfaces).**: Primitive spaces are standard Borel; histories and continuation availability are specified.
- A1 (positive experiment).**: Preparation and transition–observation kernels are nonnegative and normalized as in (2.1).
- A2 (causality).**: The next action uses only the visible past and permitted independent randomness. Stopping is nonanticipative.
- A3 (executable tests).**: The designated future tests have actual protocols and are closed under the one-step extensions needed for prediction.
- A4 (preparation and accounting).**: Average risks and acquisition laws refer to a declared preparation and exploration strategy. Failures, pilots, and accepted-event probabilities are not removed from their costs.
- A5 (charged implementation).**: Persistent labels, numerical precision, workspace, read-only calibration, and simulator state have separate accounts.

In particular these axioms do not assert strict positivity of each report, finite-dimensionality, stationarity, differentiability, a spectral gap, or a large-deviation principle. If the physical state is not Markov in the coordinates initially proposed, its entire necessary past can be included in X_t . This does not prove closure for a smaller observable state.

2.2. Strategies, paths, and stopping. A strategy is a sequence of measurable kernels $\alpha_t(da | h_t)$ supported on available actions. It cannot depend on the unknown true parameter. A finite-state strategy may have an additional explicit internal state; an unrestricted history strategy need not satisfy a finite-memory budget.

Proposition 2.2 (Path construction). *For every parameter and causal strategy there is a unique consistent finite-dimensional path law with iterated-kernel expression*

$$\mu_\lambda(dx_0) \prod_{t=0}^{N-1} \alpha_t(da_t | h_t) K_t^\lambda(dx_{t+1}, dy_{t+1}, dc_{t+1} | x_t, a_t). \quad (2.2)$$

For countably many discrete times these laws extend uniquely to the product path sigma field. Deterministic evolution and deterministic observations are included by taking Dirac kernels.

Proof. For finite N , integrate bounded measurable cylinder functions from the last variable backwards. At each step measurability follows from the kernel property, positivity is preserved, and the integral of one is one. The resulting probability measures have consistent marginals because the last kernel integrates to one. The standard kernel-extension theorem gives the countable product law; this is the Ionescu–Tulcea construction, as treated in [4]. Uniqueness follows first on cylinder events and then on their generated sigma field. A Dirac kernel makes the corresponding integral substitution along the deterministic trajectory, without changing the remaining preparation or measurement randomness. $\square$

A bounded stopping time is implemented by an absorbing stopped symbol and zero subsequent increments. Thus the same construction handles optional termination. For an unbounded stopping time, an almost-surely finite stopped path can be considered by truncation; quantitative infinite-horizon error bounds still require a summable bound or a separate limiting argument. A random number of collisions is not silently substituted for a deterministic large-deviation speed.

2.3. Prepared and parameter-uniform questions. For statistical decision theory we retain the whole family $(P_\lambda^\alpha)_\lambda$. A decision rule is one rule, independent of the true parameter. For Bayesian prediction we instead specify a prior $\eta(d\lambda)$ and use the joint hidden state (λ, x_t) with preparation $\eta(d\lambda)\mu_\lambda(dx_0)$. Fixed known calibration is permitted, but it is not a substitute for an unknown parameter.

This distinction also applies to infima over encoders. The infimum for a fixed preparation need not be achieved by the same encoder for all priors. Likewise, a family of codebooks indexed by a known experimental calibration is not an estimator of an unknown calibration. Every quantitative theorem below states which situation is intended.

2.4. Instruments and evidence. After a preparation has been fixed, write the hidden state simply as x . For a report event $B \subset Y_{t+1}$, define the positive instrument

$$(\mathbb{T}_{t,a,B}f)(x) = \int \mathbf{1}_B(y)f(x')K_t(dx', dy \mid x, a).$$

It is countably additive in B for nonnegative f , positive in f , and $\mathbb{T}_{t,a,Y}1 = 1$. With finite reports write $\mathbb{T}_{t,a,y} = \mathbb{T}_{t,a,\{y\}}$. Starting with $b_0 = \mu$, define the unnormalized history functional by

$$b_{t+1}(f) = b_t(\mathbb{T}_{t,a_t,y_{t+1}}f), \quad Z_t = b_t(1). \quad (2.3)$$

These functionals do not include the probability of the chosen command. They are likelihood/evidence functionals conditional on the realized command sequence; the actual joint history law also includes the strategy kernels in (2.2).

Proposition 2.3 (Normalized update). *For finite reports and $Z_t > 0$, let $\pi_t = b_t/Z_t$. If $q_t(y \mid \pi_t, a) = \pi_t(\mathbb{T}_{t,a,y}1) > 0$, then*

$$\pi_{t+1}(f) = \frac{\pi_t(\mathbb{T}_{t,a,y}f)}{\pi_t(\mathbb{T}_{t,a,y}1)}, \quad Z_{t+1} = Z_t q_t(y \mid \pi_t, a). \quad (2.4)$$

The formula remains valid for adaptive actions. When the evidence is zero, the unnormalized functional is zero and no normalized posterior is specified by this formula.

Proof. Divide (2.3) by its value at one. Positivity shows that a positive functional of total mass zero vanishes on every bounded test. For adaptive actions, condition on the visible history: the strategy's probability of the current action is fixed by that history and is independent of the hidden state conditional on it. This factor cancels in the posterior. An induction in (2.2) verifies the conditional expectation identity for every bounded hidden-state test. $\square$

For a non-atomic report space, conditioning is by disintegration of the joint next-state/report law, not by dividing the mass of a singleton. Under a fixed preparation a standard Borel conditional distribution exists almost everywhere. A single Borel update jointly in all proposed states, inputs, and reports is additional implementation data, or must be justified by an applicable parametrized disintegration theorem. We will not use a simultaneous version across a nondominated parameter family without that justification. All explicit quantitative models below have finite reports and Borel formulas.

2.5. Preparing a conditional experiment. If an event has acceptance probability $p > 0$, independently repeating a preparation until acceptance implements the conditional law. Its preparation count T satisfies $\mathbb{P}(T > n) = (1-p)^n$ and $\mathbb{E}T = 1/p$. A cap B instead creates an abort symbol with probability $(1-p)^B$. This is a different experiment from exact conditioning. The cap is necessary for a uniform pathwise preparation budget. No lower bound or implementation claim below replaces an actual report word by a probability-one conditioning event.

3. FUTURE TESTS AND PREDICTIVE QUOTIENTS

3.1. Operator words and adaptive protocols. A finite future protocol starts from the present time, uses available actions causally, and returns a bounded terminal readout. Its conditional expected readout, as a function of the current hidden state, is a future test. A terminal hidden-state probe is allowed only if an actual measurement kernel producing it is part of the protocol. For finite report alphabets, a deterministic depth- m protocol is a tree of actions and terminal readouts. Its test has the recursion

$$f = \sum_y \mathsf{T}_{t,a,y} f_y. \quad (3.1)$$

Randomized protocols integrate such recursions. Let $\mathcal{V}_t$ be the real linear space generated by the designated future tests, including constants. We require the one-step closure

$$\mathsf{T}_{t,a,y} \mathcal{V}_{t+1} \subset \mathcal{V}_t \quad (3.2)$$

for the continuations being retained. Where full future laws are intended, the tests include indicator readouts from a determining event class and its needed bounded-monotone closure.

A fixed finite menu used to assess prediction loss can be much smaller than $\mathcal{V}_t$. Such a menu does not automatically define a recursively sufficient state. One must either prove its closure/completeness or keep a larger predictive state for the online construction.

Definition 3.1 (Predictive equivalence). Two prepared beliefs at time t are predictively equivalent when they have the same continuation interface and

$$\pi \equiv_t \tilde{\pi} \iff \pi(f) = \tilde{\pi}(f) \quad (f \in \mathcal{V}_t).$$

The continuation interface includes the legal future action tree, remaining budget, and any public clock that affects future tests. The predictive quotient is the set of these equivalence classes, or equivalently the image of $\pi \mapsto (\pi(f))_{f \in \mathcal{V}_t}$ together with that interface.

Theorem 3.2 (Minimal future prediction). *Under (3.2), predictively equivalent beliefs give the same laws for all designated future protocols. After an action and a report of positive probability, their updated beliefs remain predictively equivalent. Conversely, equality of those protocol laws implies equality of their future tests. If a deterministic history statistic r_t determines all these predictions and the continuation interface, the predictive state factors uniquely through r_t on its attained image.*

Proof. For a one-step report, its probability is the value of $\mathsf{T}_{t,a,y} 1 \in \mathcal{V}_t$, and is therefore common. For a next-state test f , both the numerator and denominator of (2.4) have the same values under equivalent beliefs. The conditional next predictions are equal. Induct over the depth of a protocol tree using (3.1). Randomized action choices use the same conditional randomization at corresponding visible histories and can be integrated. Bounded monotone convergence extends equality from finite determining tests to the designated generated events. Conversely, expectations of each designated bounded readout are determined by its law, hence agree; equality extends linearly to $\mathcal{V}_t$.

If $r_t(h) = r_t(\tilde{h})$, sufficiency of the statistic for these tasks implies equality of all future predictions and interfaces. Thus each fibre of r_t is contained in a predictive-equivalence class. Define the factor on the image of r_t by the predictive class of any history in that fibre. It is well defined and is the unique such map. This last assertion is a set-theoretic universal property. Its measurable version requires an actual measurable quotient realization, as specified below. $\square$

The theorem does not give a normalized continuation at a zero-probability report. One can extend an update arbitrarily there for a chosen measurable implementation, but such an extension is not canonical information. For worst-history statements, admissible histories and these extensions must be explicitly specified.

3.2. Safe measurable realizations. The complete posterior on a standard Borel hidden space is a safe measurable state under a fixed experiment and appropriate conditional versions. Its predictive quotient may be simpler, but not every abstract measurable quotient is standard Borel. We use the following concrete sufficient condition.

Theorem 3.3 (Compact continuous realization). *Let B_t be compact metric admissible belief domains and let $(f_{t,j})_{j \geq 1}$ be continuous bounded evaluation functions on B_t that determine predictive equivalence. Put*

$$e_t(b) = (f_{t,j}(b))_{j \geq 1}, \quad S_t = e_t(B_t),$$

after rescaling the functions into $[-1, 1]$. Suppose action spaces are compact metric, reports finite, and $F_t : B_t \times A_t \times Y_{t+1} \rightarrow B_{t+1}$ is continuous for each report and respects predictive equivalence. Then S_t is compact metrizable and the induced update $\bar{F}_t : S_t \times A_t \times Y_{t+1} \rightarrow S_{t+1}$ is continuous for each report. Every continuous evaluation constant on predictive fibres descends to a continuous function on S_t .

Proof. The countable product $[-1, 1]^{\mathbb{N}}$ is compact metrizable, and e_t is continuous. Its image is compact, hence closed and metrizable. A continuous surjection from a compact space to a Hausdorff space is a closed quotient map. The same holds for $e_t \times \text{id}_{A_t}$ because its domain is compact and its target Hausdorff. For each report, $e_{t+1} \circ F_t$ is continuous and constant on fibres of this quotient map. The definition of quotient topology therefore gives a unique continuous descended map. The final assertion is the same argument without the action factor. $\square$

If the hidden state space is compact and the belief domain is compact in weak convergence, continuous hidden-state tests give the required continuous evaluations. With continuous positive likelihoods bounded away from zero, Bayes updating is continuous. In concrete finite models this is verified directly. A countable dense set of actions can determine all finite words when the corresponding evaluations are continuous; this must be checked, not inferred from standard Borel structure alone.

For finite histories the quotient is finite and the measurable issue is trivial. For noncompact or nondominated settings we state results on an already supplied Borel predictive realization rather than claiming a universal smooth quotient.

3.3. Finite remaining-test coordinates. The following elementary result is useful when a finite representation exists, including representations with redundant coordinates.

Proposition 3.4 (Positive finite-coordinate update). *Let $\psi_t = (1, \psi_{t,1}, \dots, \psi_{t,q_t})$ be bounded tests. Suppose*

$$\mathbb{T}_{t,a,y} 1 = b_{t,a,y} \psi_t, \quad \mathbb{T}_{t,a,y} \psi_{t+1,i} = c_{t,a,y,i} \psi_t,$$

with uniformly bounded coefficient rows and jointly Borel dependence on the input. On a domain of beliefs where every relevant report probability is at least $\kappa > 0$, the coordinate vector $v = (\pi \psi_{t,i})_{i \geq 1}$ updates by

$$v'_i = \frac{c_{t,a,y,i}(1, v)^{\mathbb{T}}}{b_{t,a,y}(1, v)^{\mathbb{T}}}. \quad (3.3)$$

It is uniformly Lipschitz on segments joining such moment vectors when the same report floor holds on their belief mixtures. Linear independence of the tests is not required.

Proof. Equation (3.3) follows from (2.4). A segment between two belief moment vectors represents their convex mixtures. The denominator is a report probability under that mixture and is at least κ . The numerator, coefficient rows, and coordinate vectors are bounded. Differentiating the quotient bounds its derivative by a constant depending only on these bounds and κ . Integrate that derivative along the segment. Redundant coefficient representations have the same evaluations on actual moment vectors and their mixtures, so they give the same update there. $\square$

In the static latent-variable case the instruments are multiplication by report likelihoods. In a dynamic model they are positive transfer operators. In either case the update uses current coordinates, command, and report, not a recovered history tape. An observer who stores the entire vector v at arbitrary real precision has not yet complied with a finite-label budget; that is the problem of Sections 6 and 7.

3.4. Three different fibres. It is useful to record the types before adding geometry. Predictive fibres are fibres of $h_t \mapsto s_t(h_t)$. Statistical fibres are equivalence classes of parameters with equal laws under the specified strategy class. Feature fibres are fibres of a chosen map $\mathbf{O} : \Lambda \rightarrow Z$. Equality of features can be strictly weaker than equality of laws, and equality of a fixed-prior posterior prediction need not be parameter-uniform sufficiency. Their derivatives, when defined, have different domains.

4. EXECUTABLE REDUCTIONS AND CHARGED RESOURCES

4.1. Finite-state implementations. A synchronized causal machine between experiments has an internal state r_t . At a target step it receives the target action a'_t , chooses a source action according to a parameter-independent kernel depending on (r_t, a'_t) , receives the source report, and outputs a target report while updating r_t . Random seeds are independent of the parameter and physical preparation. A stored seed that encodes previous reports is part of r_t . Every raw source step and every retained state is charged.

This synchronized model suffices for the results below. A macro-step that uses a random number of source steps requires an explicit causal stopping protocol and its preparation certificate. Such a protocol can be included, but a bound for one synchronized source step is not silently reused for it.

A target controller with state z_t receives the target reports and updates z_t . Its joint implementation with the machine stores (r_t, z_t) . Thus $|r_t| \leq R_t$ and $|z_t| \leq M_t$ give the direct product bound $R_t M_t$. This is an upper bound for one realization, not a claim that no smaller implementation is possible.

Definition 4.1 (Causal reduction certificate). For experiments over a common parameter space, a certificate $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ consists of a parameter-independent causal machine, a lift $\mathbf{L}_\Gamma$ of every admitted target controller to its source implementation, and a monotone resource map ρ_Γ . Under target controller α , write $\hat{P}_\lambda^{\Gamma, \alpha}$ for the simulated target-history law obtained from the joint machine/source execution. An ε -certificate satisfies

$$\sup_{\lambda, \alpha} \left\| \hat{P}_\lambda^{\Gamma, \alpha} - P_{\mathfrak{F}, \lambda}^\alpha \right\|_{\text{TV}} \leq \varepsilon, \quad (4.1)$$

and implementing a target budget B costs at most $\rho_\Gamma(B)$. The default certificate is pathwise. Expected-cost and tail-budget certificates are different types and must be stated separately.

For randomized strategy lifts, the right object is the joint causal execution. It need not be representable by a terminal kernel on the raw source history alone that is independent of the controller. Once the lift and controller are fixed, its visible pushforward is unambiguous. This avoids a misleading notation that would hide feedback inside a supposedly static garbling.

An exact certificate has $\varepsilon = 0$. The identity machine gives the identity reduction. A reparametrization $r : \Lambda' \rightarrow \Lambda$ pulls back an experiment family and is kept separate from data reduction. If the reparametrization is not injective it introduces parameter redundancy; it is not loss of observed data.

4.2. The path-kernel estimate. We use the convention $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. For $0 \leq f \leq L$ this implies $|Pf - Qf| \leq L \|P - Q\|_{\text{TV}}$.

Lemma 4.2 (Predictable telescoping). *Let P and Q be laws on the same finite visible history tree, with the same initial law and the same feedback action kernels. Suppose their conditional report kernels at history h_t and action a_t are $p_t(\cdot \mid h_t, a_t)$ and $q_t(\cdot \mid h_t, a_t)$. If*

$$\|p_t(\cdot \mid h_t, a_t) - q_t(\cdot \mid h_t, a_t)\|_{\text{TV}} \leq d_t(h_t, a_t),$$

then

$$\|P - Q\|_{\text{TV}} \leq \min \left\{ 1, \mathbb{E}_P \sum_{t=0}^{N-1} d_t(H_t, A_t) \right\}. \quad (4.2)$$

If both systems become absorbing under the same visible stopping rule σ , the summand can be multiplied by $\mathbf{1}_{\{t < \sigma\}}$.

Proof. Fix a terminal measurable test $0 \leq f \leq 1$. Replace the report kernels one at a time, using a P -prefix and a Q -suffix. The suffix conditional expectation of f is a measurable function between zero and one. The change at step t , after integrating the common feedback action kernel, is therefore bounded in absolute value by $\mathbb{E}_P d_t(H_t, A_t)$. Summing the telescoping differences proves the bound for f . Take the supremum over event indicators. After the visible rule has stopped, the kernels are the same absorbing kernel and their local error is zero. Histories of zero probability under P contribute nothing; the kernels are nevertheless specified on all admitted histories so that all intermediate laws exist. $\square$

The expectation is under the actual reference experiment. In particular a rare report carrying a large conditional error is not automatically charged as if it occurred at every time. No maximal-coupling measurability theorem is needed for this proof. The action policy may be discontinuous: the same kernel is applied on the same history in each telescoping term.

Theorem 4.3 (Composition and risk transfer). *Suppose $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$ and $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$ have certificates (ε_1, ρ_1) and (ε_2, ρ_2) uniformly over the controller classes needed by the lifts. Their serial implementation has certificate*

$$(\min\{1, \varepsilon_1 + \varepsilon_2\}, \rho_1 \circ \rho_2).$$

Exact reductions compose associatively as executions, up to relabeling of product states. For a common target $\tau(\lambda)$ and bounded loss $0 \leq \ell \leq L$,

$$\mathfrak{R}_{\mathfrak{E}}(\rho_{\Gamma}(B)) \leq \mathfrak{R}_{\mathfrak{F}}(B) + L\varepsilon \quad (4.3)$$

for either minimax risk or risk under a common prior.

Proof. Execute the second machine as a controller of the first. Their stored states form a product and their source actions remain nonanticipative and parameter independent. The first certificate applies to this enlarged controller class by hypothesis. The subsequent machine/controller execution is a Markov postprocessing of its visible history together with its independent randomization, so it contracts total variation. Compare first with an exact $\mathfrak{F}$ run and then with the intended $\mathfrak{G}$ run. The triangle inequality gives the displayed defect. Budget maps compose in the same order because they map target requirements to source requirements. Regrouping three nested machine states changes only their names, proving associativity.

For a target procedure within budget B , run its source implementation and use the same final decision. Its risk at each parameter differs by at most $L\varepsilon$. The resource certificate makes it admissible under $\rho_{\Gamma}(B)$. Take the supremum over parameters, or integrate against the common prior, and then take the infimum over target procedures. An approximating sequence of procedures suffices if an optimum is not attained. $\square$

A risk certificate concerns the same decision target and loss on both sides. It does not compare excess errors relative to different full-history predictors. An expected preparation constraint cannot

be transported by a pathwise certificate without checking its type. Conversely, total-variation closeness alone does not transfer an unbounded cost expectation.

4.3. Checking an exact state reduction locally.

Proposition 4.4 (One-step realization). *Let $r_t : X_t \rightarrow X'_t$, $g_t : Y_t \rightarrow Y'_t$, and an action lift $\iota_t : A'_t \rightarrow A_t$ be measurable. Suppose the preparations push forward correctly and, for every admitted source state and target action,*

$$(r_{t+1}, g_{t+1})_{\#} K_t^{\lambda}(\cdot \mid x, \iota_t a') = K_t^{\lambda}(\cdot \mid r_t x, a'). \quad (4.4)$$

Suppose also that continuation interfaces, recorded costs, and designated readouts match. Then applying the lifted actions and report maps gives an exact causal reduction for every target feedback strategy.

Proof. At time zero the reduced state laws agree. Conditional on a source state and on a lifted action, (4.4) gives the joint reduced next-state/report law. Integrate against the common current reduced law. Since the target controller reads the reduced history, its next action law is the same in both systems. Induction proves equality on all finite cylinders, and hence on the path sigma field. Readout and cost matching give the asserted experiment and resource identity. $\square$

The state map need not be observable to implement the reduction: the machine only needs the action lift and report map. The hidden-state map is a proof device for the law identity. If the proposed lift instead requires the unobserved state, it is not an executable reduction of this type.

4.4. Terminal and causal equivalence. For a fixed terminal family, a garbling G is reversible if there exists a parameter-independent kernel R such that $P_{\lambda} G R = P_{\lambda}$. This is statistical equivalence. Requiring R to operate causally and charging its memory gives a stronger notion. Requiring suitable bounds in both directions gives resource-aware causal equivalence. These notions are not identified by definition.

Independent parallel experiments may be implemented by product machines when their preparations and physical randomness are genuinely independent conditional on the parameter. Persistent cardinalities multiply and costs add. Shared nuisance variables or coupled preparations require the joint experiment; replacing them by an independent product changes the object.

5. POSTERIOR GEOMETRY AND INFORMATION LOSS

5.1. Posterior barycentres. The following statements concern a common preparation. Let Z be a hidden target, Y an original observation, and W a parameter-independent postprocessing of Y . Conditional independence is denoted by the Markov chain $Z \rightarrow Y \rightarrow W$.

Theorem 5.1 (Posterior barycentre and convex order). *For bounded measurable $f_1, \dots, f_q$ of Z , put*

$$p_Y = (\mathbb{E}[f_i(Z) \mid Y])_i, \quad p_W = (\mathbb{E}[f_i(Z) \mid W])_i.$$

Then

$$p_W = \mathbb{E}[p_Y \mid W]. \quad (5.1)$$

Consequently $\mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$ for every convex function for which the expectations are defined. Moreover,

$$\text{Cov}(p_Y) - \text{Cov}(p_W) = \mathbb{E} \text{Cov}(p_Y \mid W) \succeq 0. \quad (5.2)$$

Proof. The Markov property gives $\mathbb{E}[f_i(Z) \mid Y, W] = \mathbb{E}[f_i(Z) \mid Y]$. Taking conditional expectation with respect to W proves (5.1). Conditional Jensen yields the convex-order inequality. Subtract the common mean and expand the orthogonal decomposition $p_Y - \mathbb{E}p_Y = (p_Y - p_W) + (p_W - \mathbb{E}p_W)$; its cross covariance is zero by conditioning on W . This gives (5.2). $\square$

For a measure-valued posterior, the same identity holds against every bounded test. It is a conditional barycentre, not a function sending each original posterior to a single coarser posterior. In particular the image of a finite set of original posteriors may generate a continuous family of conditional barycentres. This is why convex-order monotonicity is not a statement about the cardinality of a predictive representation.

5.2. Relative entropy and recovery. For $P \ll Q$, set $D(P\|Q) = \int \log(dP/dQ) dP$, allowing $+\infty$; set it to infinity otherwise. Let $G : X \rightsquigarrow W$ be a Markov kernel and form the joint laws $\bar{P}(dx, dw) = P(dx)G(dw | x)$ and $\bar{Q}(dx, dw) = Q(dx)G(dw | x)$.

Theorem 5.2 (Entropy loss and a common recovery kernel). *If $D(P\|Q) < \infty$, then*

$$D(P\|Q) = D(PG\|QG) + \int D(\bar{P}(dx | w)\|\bar{Q}(dx | w)) PG(dw). \quad (5.3)$$

In particular, entropy decreases under G . Suppose a family P_λ is dominated by one P_0 , every $D(P_\lambda\|P_0)$ is finite, and

$$D(P_\lambda G\|P_0 G) = D(P_\lambda\|P_0) \quad \text{for every } \lambda.$$

Then $R(w, dx) = \bar{P}_0(dx | w)$ is a parameter-independent recovery kernel satisfying $P_\lambda GR = P_\lambda$ for every λ .

Proof. The joint likelihood ratio is $d\bar{P}/d\bar{Q}(x, w) = dP/dQ(x)$. Disintegrating both joint laws gives the factorization into the marginal likelihood ratio on W and the conditional likelihood ratio on X . Take logarithms and integrate. Standard truncation of the densities justifies the formula when logarithms are not bounded; finite total entropy makes the nonnegative conditional entropy integrable. This proves (5.3). Alternatively the marginal ratio is the conditional expectation under $\bar{Q}$ of dP/dQ , and conditional Jensen proves the inequality directly, also for infinite entropy.

For the family, equality makes the conditional entropy vanish $P_\lambda G$ -almost everywhere. Hence the two conditional distributions coincide there. The conditional law under the one reference P_0 is the same measurable kernel for all parameters; $P_\lambda G \ll P_0 G$ handles its null-set convention. Integrating that conditional law against $P_\lambda G$ recovers P_λ . $\square$

This is a sufficient recovery criterion under a common reference. An isolated equality for one pair does not certify an unrelated parameter family. Nor does it make the recovery kernel causal or memory free.

5.3. Scores and Fisher information. To avoid hiding regularity in notation, first take a finite observation space with $p_\lambda(y) > 0$ and continuously differentiable probabilities on an open parameter domain in $\mathbb{R}^k$. Let

$$s_\lambda(y) = \nabla_\lambda \log p_\lambda(y), \quad \mathcal{I}_\lambda = \mathbb{E}_\lambda[s_\lambda s_\lambda^\top].$$

If $G(w | y)$ is a parameter-independent finite channel, restrict to output symbols having positive probability; identically impossible symbols can be removed.

Theorem 5.3 (Score projection). *The processed score and information satisfy*

$$s_\lambda^G(W) = \mathbb{E}_\lambda[s_\lambda(Y) | W], \quad (5.4)$$

$$\mathcal{I}_\lambda - \mathcal{I}_\lambda^G = \mathbb{E}_\lambda \text{Cov}_\lambda(s_\lambda(Y) | W) \succeq 0. \quad (5.5)$$

For a direction v , equality of information in that direction is equivalent to $v^\top s_\lambda(Y)$ being determined by W almost surely.

Proof. Differentiate $q_\lambda(w) = \sum_y p_\lambda(y)G(w | y)$ and divide by $q_\lambda(w)$. The coefficients are precisely the conditional probabilities of Y given W , proving (5.4). Differentiating $\sum_y p_\lambda(y) = 1$ gives $\mathbb{E}_\lambda s_\lambda = 0$. The conditional variance identity now gives (5.5). Its quadratic form in v is the mean conditional variance of $v^\top s_\lambda$, which vanishes exactly under the stated measurability condition. $\square$

The same proof works in a dominated differentiable model if differentiation under the integral is justified and the score lies in L^2 . More generally, the QMD version is the classical score-projection theorem described by Pollard [11]; when it is invoked, quadratic-mean differentiability is an additional statistical hypothesis. The raw experiment axioms by themselves impose no differentiability.

Information equality in (5.5) is a statement at a parameter point. It is not the recovery conclusion of Theorem 5.2. Even information preservation throughout a model need not make a statistic sufficient without stronger assumptions; see [11].

Under a smooth parameter map $\lambda = r(\zeta)$, the information pulls back as $Dr^\top \mathcal{I} Dr$. Under a deterministic linear map of a random vector, its covariance pushes forward as $A\Sigma A^\top$. These are different operations on different objects. Neither formula says that an arbitrary covariance equals inverse Fisher information.

5.4. Sequential information in finite positive models. A useful controlled specialization follows without differentiating the controller. Suppose the conditional report probabilities are finite, positive, and differentiable, and the feedback policy has no explicit unknown-parameter dependence. Along a history define the local score

$$\xi_{t+1} = \nabla_\lambda \log p_\lambda(Y_{t+1} \mid H_t, A_t).$$

The policy factors contribute no score. Conditional normalization gives $\mathbb{E}_\lambda[\xi_{t+1} \mid H_t, A_t] = 0$.

Proposition 5.4 (Sequential score decomposition). *If the local scores are square integrable, the path score is $S_N = \sum_{t < N} \xi_{t+1}$, and*

$$\mathbb{E}_\lambda[S_N S_N^\top] = \sum_{t < N} \mathbb{E}_\lambda[\xi_{t+1} \xi_{t+1}^\top]. \quad (5.6)$$

For bounded stopping, each term is restricted to active steps. No independence of successive reports is required.

Proof. The likelihood factors into conditional report probabilities and parameter-independent policy factors. Its logarithmic derivative is the displayed sum. For $s < t$, ξ_{s+1} is measurable before ξ_{t+1} , so their cross moment is zero by the conditional mean-zero property. Expanding the square proves (5.6). A bounded stopping indicator is predictable for the next increment; the same argument applies. $\square$

This is an information identity for the actual controlled law. It does not make an information-optimal strategy automatically prediction-memory optimal, and it does not supply an asymptotic normal approximation.

6. ATTAINED PREDICTION AND CHECKPOINT QUANTIZATION

6.1. The physical loss. Fix a preparation and an exploration law, independently of coding randomness. At time t choose a future query J independently after the label is formed, with probabilities $w_j > 0$, $\sum_j w_j = 1$. Conditional on history and query, a bounded response Z_j has mean $m_j(h)$. Squared loss is used, and only the selected query is executed. Set

$$p(h) = (\sqrt{w_j} m_j(h))_{j=1}^q, \quad \nu = \text{Law}(p(H_t)).$$

The weighted decoder vector is denoted by c . Subtracting the complete history predictor's risk leaves $\|p(h) - c\|^2$.

An M -label checkpoint encoder reads the complete prefix once and retains one of M symbols. Its decoder receives the label and the subsequent query, not the discarded history or the acquisition commands. A public seed may specify a codebook but is independent of the preparation. Private coins are allowed and are either discarded or included in the retained label. We assume a bounded

closed convex prediction domain containing all $p(h)$; projection into it can only improve squared error.

Theorem 6.1 (Exact checkpoint identity). *The optimal mean checkpoint excess risk equals*

$$R_M^{\text{chk}} = e_M^2(\nu) := \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp). \quad (6.1)$$

Independent public or private coding randomization does not decrease this infimum. An arbitrarily near-optimal deterministic nearest-centre encoder is measurable.

Proof. Condition on history, query, and the code randomness. The elementary identity

$$\mathbb{E}[(Z_j - a)^2 \mid h, J = j] = \text{Var}(Z_j \mid h, J = j) + (m_j(h) - a)^2$$

reduces the loss to the displayed Euclidean distortion. For each label, replace a randomized decoder by its conditional mean over decoder coins; convexity of squared norm does not increase the loss. Given fixed decoder centres, random assignment by the encoder is no better than the nearest centre. Hence every fixed public-seed codebook has risk at least the right-hand side of (6.1); averaging over the independent seed preserves this inequality.

Conversely choose centres within any prescribed positive tolerance of the infimum. Encode by the first index attaining the least distance. Finite comparisons of continuous distance functions make this rule Borel. The resulting distortion is the integrand in (6.1). Let the tolerance decrease to zero. A codebook indexed by an unknown true parameter is not produced by this argument: ν is the one declared preparation law. $\square$

For worst-history loss the analogous deterministic checkpoint quantity is a covering radius, not the integral in (6.1). The mean and worst-history laws have the same order only under additional geometric and mass conditions.

6.2. A small-ball lower bound.

Lemma 6.2 (Mass obstruction). *Suppose a probability law ν in $\mathbb{R}^d$ satisfies $\nu(B(c, r)) \leq Ar^d$ for every centre c and radius $r > 0$. Then for every integer $M \geq 1$,*

$$e_M^2(\nu) \geq \frac{1}{2}(2AM)^{-2/d}. \quad (6.2)$$

In particular, a Lebesgue density bounded by $B\epsilon^{-d}$ gives $e_M^2(\nu) \geq c_{d,B}\epsilon^2 M^{-2/d}$ for $\epsilon > 0$.

Proof. The union of M balls of radius $r = (2AM)^{-1/d}$ has probability at most one half. On its complement every centre is at least distance r away. Integrating the nearest-centre squared distance gives at least $r^2/2$. For the density assertion use $A = Bv_d\epsilon^{-d}$, where v_d is the volume of the Euclidean unit ball. $\square$

The lemma permits centres outside the support and therefore already covers arbitrary decoders. It does not condition away a small probability of reaching the difficult region. If the useful density lower geometry instead occurs only on an event of probability β , the eventual risk lower bound must carry that mass.

Lemma 6.3 (A positive attained chart). *Suppose a coordinate projection of a prediction law dominates β times uniform probability on a d -dimensional cube of side a , with $\beta, a > 0$. If that projection has operator norm at most L , then*

$$e_M^2(\nu) \geq c_d \beta a^2 L^{-2} M^{-2/d}.$$

Proof. For any centres in the original prediction space, project the centres onto the cube's coordinate space. Distances there are at most L times the original distances. The uniform cube law has density a^{-d} and satisfies Lemma 6.2. Domination by β and rescaling prove the assertion. $\square$

6.3. Whole-image covers and all integer budgets. Let $D \subset \mathbb{R}^d$ be nonempty, compact, and contained in a cube of side $2a$. Divide each coordinate interval into $k = \lfloor M^{1/d} \rfloor$ pieces. There are at most $k^d \leq M$ cells. Choose one point of D in each nonempty cell. Any point is within Euclidean distance $2a\sqrt{d}/k$ of its representative. Since $k \geq M^{1/d}/2$, we obtain a radius $4a\sqrt{d}M^{-1/d}$ for every integer $M \geq 1$. The corresponding ℓ^1 bound is $4adM^{-1/d}$. Neither a large-budget asymptotic nor an integer-budget relaxation is needed.

More generally, finitely many Lipschitz images of compact cubes have such covers with exponent equal to the largest domain dimension. Allocate centres between the finitely many charts, use the preceding grid, and enlarge the constant for budgets smaller than the number of charts. This is a sufficient global description, not a consequence of local rank alone. A nonlinear or disconnected image cannot be replaced by a tangent plane without checking its entire cover.

6.4. Causal lower bounds use one exploration law. Let $\mathcal{C}_M$ be causal encoders with at most M_t labels at time t . Under the fixed exploration strategy, their label is also a measurable, possibly randomized, encoding of the complete time- t prefix. Therefore

$$\inf_{C \in \mathcal{C}_M} \max_{1 \leq t \leq N} \mathbb{E} \ell_t^C \geq \max_{1 \leq t \leq N} e_{M_t}^2(\nu_t). \quad (6.3)$$

Indeed the checkpoint lower inequality holds for every single C and all t before taking the infimum. The same law of exploration supplies all marginals ν_t . Changing the exploration strategy with the encoder, or with the checkpoint after the infimum, would define another problem.

6.5. Calibrated anisotropy. A useful model interface is provided by [15]. There, under whole-image covering, positive attained mass for each initial scale flag, and uniformly stable raw updates, the scale profile is

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left(\frac{s_{t,1} \cdots s_{t,\ell}}{M} \right)^{2/\ell}.$$

The physical prediction metric is fixed; small singular values are not removed by an unbounded coordinate whitening. The finite-horizon causal bound follows by propagating the covering radii, as in the next section. We cite this refined profile as a conditional model interface, not as an unproved universal consequence of a list of singular values. The dynamic model of Section 8 will establish its own isotropic scale, density bound, and invariant cover directly from the experiment.

7. CAUSAL QUANTIZATION WITH FEEDBACK AND STOPPING

7.1. Admissible state domains. Consider a predictive realization $s_t(h_t)$ with a known initial state s_0 . Let (D_t, d_t) be admissible metric state domains containing every attainable state. They may also contain normalized beliefs that are not attained at that precise time. This causes no problem if all updates and readouts are genuinely defined on D_t . What is forbidden is evaluating an update at an artificial point where it has no experimental meaning or no stated extension.

Assume common action availability on each domain, or work within a fixed continuation-interface component. For each input (a, y) , let $F_t(\cdot, a, y) : D_t \rightarrow D_{t+1}$ be Borel and satisfy

$$d_{t+1}(F_t(s, a, y), F_t(\tilde{s}, a, y)) \leq L_t d_t(s, \tilde{s}). \quad (7.1)$$

The known-calibration formulas implementing F_t are independent of the unknown parameter. If there is a parameterized law, write the next-report kernel as $g_t^\lambda(dy \mid s, a)$ and assume

$$\left\| g_t^\lambda(\cdot \mid s, a) - g_t^\lambda(\cdot \mid \tilde{s}, a) \right\|_{\text{TV}} \leq \kappa_t d_t(s, \tilde{s}) \quad (7.2)$$

uniformly over the relevant parameter domain. The same Borel state update must represent every law for a parameter-uniform statement. For a single prepared Bayesian experiment there is no additional parameter supremum.

Choose finite representative sets $C_t \subset D_t$ with $|C_t| \leq M_t$ and Borel quantizers $q_t : D_t \rightarrow C_t$ satisfying $d_t(s, q_t s) \leq r_t$. A numerical update $\tilde{F}_t(c, a, y) \in D_{t+1}$ may replace the exact update at a representative, with

$$d_{t+1}(\tilde{F}_t(c, a, y), F_t(c, a, y)) \leq \eta_{t+1}. \quad (7.3)$$

Set $\hat{s}_0 = s_0$ and implement

$$\hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (7.4)$$

The persistent state is only the index of $\hat{s}_t$ in C_t .

Theorem 7.1 (Finite-state emulation). *Under (7.1)–(7.3), define*

$$b_0 = 0, \quad b_{t+1} = L_t b_t + r_{t+1} + \eta_{t+1}, \quad b_t = \sum_{j=1}^t (r_j + \eta_j) \prod_{i=j}^{t-1} L_i. \quad (7.5)$$

Then the following assertions hold.

(i) *On every admitted common command/report sequence, $d_t(s_t, \hat{s}_t) \leq b_t$. If the physical prediction readout is K_t -Lipschitz, its excess squared error is at most $K_t^2 b_t^2$.*

(ii) *Define the finite-state reference experiment to use the same representative update (7.4), but to draw its next report from $g_t^\lambda(\cdot \mid \hat{s}_t, a_t)$. For any common causal feedback controller, the visible path laws of the actual experiment with the representative machine and of this finite-state reference experiment satisfy*

$$\left\| P_{N,\lambda}^\alpha - \hat{P}_{N,\lambda}^\alpha \right\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=0}^{N-1} \kappa_t b_t \right\}. \quad (7.6)$$

The histories here include the applied actions, reports, and retained labels. A shared visible stopping rule replaces the deterministic sum by $\mathbb{E}_{P_\lambda^\alpha} \sum_{t < \sigma \wedge N} \kappa_t b_t$.

(iii) *Any additional causal simulator is charged as in Theorem 4.3. A feedback controller with D_t own states is implemented with at most $D_t M_t$ persistent pairs. For a common bounded path-dependent loss, the risk-transfer error is at most its range times the right side of (7.6).*

Proof. For (i), insert successively the exact update from the retained representative and its numerical approximation. The triangle inequality, (7.1), the numerical error, and the covering radius give

$$d_{t+1}(s_{t+1}, \hat{s}_{t+1}) \leq L_t d_t(s_t, \hat{s}_t) + \eta_{t+1} + r_{t+1}.$$

Induction proves (7.5). The physical readout bound follows by its Lipschitz property and the conditional squared-loss identity. This argument remains valid when the actions depend on the retained label: it compares the exact posterior and the representative along the actual same sequence of applied commands and reports.

For (ii), at any common visible prefix the applied controller has the same action kernel. The representative $\hat{s}_t$ is the same measurable function of that prefix in both experiments. In the actual law, the exact predictive state is s_t ; in the finite-state reference law, the next-report kernel is evaluated at $\hat{s}_t$. Their conditional variation distance is at most $\kappa_t b_t$ by (i) and (7.2). The next label is a common deterministic function of the action, report, and current label, so appending it does not increase the conditional variation distance. Apply Lemma 4.2. Stopping makes the conditional kernels identical after the stopping time, which gives the predictable active-step bound.

For (iii), pair the retained representative with the controller's internal state. Each next action and update then reads only this pair and current inputs. The state cardinality is bounded by the product. The bounded-loss and extra-simulator statements are precisely the conclusions of Theorem 4.3. $\square$

The theorem is a directional emulation statement. It does not construct a reverse kernel from the finite-state reference experiment to the raw physical data. It therefore does not claim two-sided

Le Cam equivalence. Nor does it estimate an unknown predictive update. The implemented update and representatives are known; uniformity over a statistical family is available only under the explicitly common update conditions above.

7.2. What feedback does and does not require. There are two different uses of a finite filter. One can predict on an arbitrarily selected actual input sequence; this uses only Theorem 7.1(i). One can also use a compressed controller to choose inputs. That still gives accurate filtering along its own actual run, but comparison with the original full-information optimal controller is another problem. Part (ii) compares a common controller across two experiments and so supplies a valid route through experiment simulation. It does not require the controller to be Lipschitz, because identical prefixes produce identical action laws. A discontinuous policy evaluated at two different approximate states is not being assumed to choose the same action.

For example, minimizing a terminal common decision loss over all D_t -state controllers in the finite-state reference experiment gives a source controller with $D_t M_t$ states and the corresponding (7.6) risk allowance. This need not approximate the unrestricted optimal source policy, whose comparison would require a separate richness theorem for the controller class.

7.3. Uniform prediction and finite-horizon path comparison.

Corollary 7.2 (Contractive implementation). *If $L_t \leq \rho < 1$, $r_t \leq r$, and $\eta_t \leq \eta$ uniformly, then*

$$\sup_t d_t(s_t, \hat{s}_t) \leq \frac{r + \eta}{1 - \rho}.$$

For a bounded physical readout with uniform Lipschitz constant K , the prediction excess is at most $K^2(r + \eta)^2/(1 - \rho)^2$ at every time. If also $\kappa_t \leq \kappa$, the path-law error on N steps is at most $\min\{1, N\kappa(r + \eta)/(1 - \rho)\}$.

Proof. Sum the geometric series in (7.5), then apply the two parts of Theorem 7.1. $\square$

A uniform per-checkpoint prediction bound is not uniform total-variation closeness of two infinite path laws. The last display may grow with N . Two processes that are arbitrarily close at each step can remain statistically distinguishable from a sufficiently long record. If a general construction uses a different arbitrary codebook at each time, it is a time-indexed machine description, not automatically a finite uniform program. The dynamic construction in the next section uses one fixed invariant domain and one fixed codebook, so that distinction does not obstruct its infinite-time statement.

7.4. Model or calibration perturbations. Suppose an implementable reference update F_t^0 has $d(F_t(s, a, y), F_t^0(s, a, y)) \leq \delta_{t+1}$ on the relevant domain, and its report law differs by at most ζ_t at the same state. Repeating the proof replaces $r_j + \eta_j$ by $r_j + \eta_j + \delta_j$ in (7.5) and $\kappa_t b_t$ by $\kappa_t b_t + \zeta_t$ in (7.6). This is a deterministic certificate, not an unpaid pilot. If a preliminary estimate yields such bounds only on an event of probability $1 - p$, bounded risks receive an additional allowance of at most their range times p , and the pilot's data and persistent state must be charged.

The perturbation must remain in the same physical metric and admissible domain. A reparametrization with an unbounded inverse near a degeneracy cannot be inserted while keeping the old constants.

8. A DYNAMIC MODEL WITH A UNIFORM MEMORY LAW

8.1. The experiment and its resource convention. Fix $d \geq 1$, let $E_d = \{0, 1, \dots, d\}$, and write $\nu_i = 1/(d + 1)$. The known parameters satisfy

$$4/5 \leq \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (8.1)$$

Prepare $X_0 \sim \nu$. At each acquisition step the hidden state is updated by the transition matrix

$$P_\gamma(i, j) = (1 - \gamma)\mathbf{1}_{\{i=j\}} + \gamma\nu_j. \quad (8.2)$$

A command $u_t \in [-1, 1]^d$, chosen before the report, gives a binary report $Y_t \in \{-1, 1\}$ with likelihoods

$$\ell_y(u, i) = \begin{cases} 1/2, & i = 0, \\ (1 + y\epsilon \tanh u_i)/2, & 1 \leq i \leq d. \end{cases} \quad (8.3)$$

Conditional on the hidden path and commands, the report randomizations are independent. The resulting joint transition/report kernel is $P_\gamma(i, j)\ell_y(u, j)$. It is an instance of Definition 2.1; all report probabilities lie between $3/8$ and $5/8$.

For the acquisition lower bound the commands are independent uniform on $[-1, 1]^d$, independent of the hidden process and coding randomness. The upper-bound filter will work on every admitted command/report sequence, including adaptively chosen commands. It does not change the fixed exploration law used for the lower bound.

At any selected time $t \geq 1$ the acquisition can stop and one terminal probe is executed without an intervening refresh. After the retained label is formed, draw J uniformly from $\{1, \dots, d\}$. Probe j produces $Z \in \{0, 1\}$ with

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\mathbf{1}_{\{i=j\}}, \quad \tau = 1/4. \quad (8.4)$$

This is one noisy binary readout, not observation of X_t or of the full posterior. Its choice is independent of the acquisition. Each acquisition and the terminal probe costs one recorded experimental step. Evaluating maximum-checkpoint risk means that a run executes only its selected probe; it does not execute counterfactual probes at all times.

Let $p_t = \text{Law}(X_t \mid H_t)$, and write $p_{t,+} = (p_{t,1}, \dots, p_{t,d})$. The probe's full-history success probability is $1/2 + \tau p_{t,j}$. If a decoder uses candidate coordinates c_j , its excess loss is

$$\frac{\tau^2}{d} \|p_{t,+} - c\|_2^2. \quad (8.5)$$

Arbitrary probe predictions are allowed; projecting them onto the interval $[1/2, 1/2 + \tau]$ cannot worsen loss. Thus the decoder coordinate formulation loses no candidates for the infimum.

Let $R_{M,t}^{\text{av}}$ and $R_{M,t}^{\text{max}}$ be optimal checkpoint risks under the stated exploration and over all admitted histories, respectively. Let $\mathfrak{R}_{M,N}^{\text{av}}$ and $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding best maximum risks for one causal M -label encoder at times $1, \dots, N$. The clock, d, γ, ϵ , and the finite read-only program are known; no history or unknown calibration is stored in those constants. At time zero one fixed initial state is allowed. For $N = \infty$ replace maxima by suprema and use one machine for all times.

Theorem 8.1 (Refreshing experiment memory law). *For every fixed $d \geq 1$, constants $0 < c_d \leq C_d < \infty$ exist such that, for every (8.1), every integer $M \geq 1$, every $N \in \mathbb{N} \cup \{\infty\}$, and every $1 \leq t \leq N$,*

$$c_d \epsilon^2 M^{-2/d} \leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}, \quad (8.6)$$

$$c_d \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,N}^{\text{av}} \leq \mathfrak{R}_{M,N}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}. \quad (8.7)$$

The upper bounds are realized by one deterministic finite-label filter using a fixed codebook for all positive times. The lower bounds allow independent public and private coding randomization. At $\epsilon = 0$ all four risks are zero with one label. Constants are uniform in time, horizon, calibration within (8.1), and the integer budget.

The proof will derive the cover, the actual attained density, and the stable update separately. This also supplies a concrete dynamic realization of the general emulation theorem.

8.2. Explicit posterior recursion. Set $p_0 = \nu$. Given the previous posterior p , the distribution just before the next report is

$$q = (1 - \gamma)p + \gamma\nu, \quad q_i \geq m_* := \frac{4}{5(d+1)}. \quad (8.8)$$

For a command and report put

$$r_0 = 1, \quad r_i = 1 + y\epsilon \tanh u_i \quad (i \geq 1), \quad D_y = q_0 + \sum_{i=1}^d q_i r_i.$$

Then the probability of that report is $D_y/2$, and

$$F(p, u, y)_0 = \frac{q_0}{D_y}, \quad F(p, u, y)_i = \frac{q_i r_i}{D_y} \quad (i \geq 1). \quad (8.9)$$

Every denominator belongs to $[1 - \epsilon, 1 + \epsilon]$. These formulas hold for every belief in the full simplex and every admitted input. Predictable actions reveal no further hidden-state information once their past is conditioned upon, as in Proposition 2.3.

Lemma 8.2 (Uniform contraction). *For any two simplex beliefs and the same (u, y) ,*

$$\|F(p, u, y) - F(\tilde{p}, u, y)\|_1 \leq \rho \|p - \tilde{p}\|_1, \quad \rho = 2 \frac{1 + \epsilon}{1 - \epsilon} (1 - \gamma) \leq \frac{2}{3}. \quad (8.10)$$

Proof. For fixed positive likelihood vector r , write $B_r(q) = \text{diag}(r)q/(q^T r)$. Subtract the two normalized vectors by adding and subtracting $\text{diag}(r)\tilde{q}/(q^T r)$. The first term has norm at most $(r_{\max}/r_{\min}) \|q - \tilde{q}\|_1$. The second has norm $|q^T r - \tilde{q}^T r|/(q^T r)$, at most the same bound. Thus $\text{Lip}_1(B_r) \leq 2r_{\max}/r_{\min}$. Equation (8.8) multiplies differences by $1 - \gamma$. Use $r_{\max} \leq 1 + \epsilon$, $r_{\min} \geq 1 - \epsilon$, and (8.1) to obtain the result. $\square$

Lemma 8.3 (An invariant shrinking domain). *Let*

$$a_\epsilon = \frac{2\epsilon}{\gamma(1 - \epsilon)}, \quad D_\epsilon = \{p \in \Delta_d : \|p - \nu\|_1 \leq a_\epsilon\}. \quad (8.11)$$

Then $\nu \in D_\epsilon$, $F(D_\epsilon, u, y) \subset D_\epsilon$, and $a_\epsilon \leq (10/3)\epsilon$. Every attainable posterior belongs to D_ϵ , for every time and every admitted history.

Proof. For any pre-report belief q ,

$$\|B_r(q) - q\|_1 = \sum_i q_i \frac{|r_i - q^T r|}{q^T r} \leq \frac{2\epsilon}{1 - \epsilon}.$$

Consequently

$$\|F(p, u, y) - \nu\|_1 \leq (1 - \gamma) \|p - \nu\|_1 + \frac{2\epsilon}{1 - \epsilon}.$$

The right side is at most a_ϵ when $p \in D_\epsilon$, by its definition. The numerical bound follows from $\gamma \geq 4/5$ and $1 - \epsilon \geq 3/4$. Start the induction at ν . $\square$

The set D_ϵ is a compact convex polytope for each fixed calibration. Its representatives need not all be attainable at a particular time; they are normalized beliefs with valid updates and belong to a proved invariant domain. No extrapolation outside the probability simplex is used. At $\epsilon = 0$ the domain is the singleton $\{\nu\}$.

8.3. The actual acquisition density. Fix a preceding history, so its pre-report belief q is fixed with (8.8). For $\epsilon > 0$ and a fixed sign y , the command map

$$\Phi_{q,y} : u \mapsto F(p, u, y)_+ \in \mathbb{R}^d$$

is one-to-one. Indeed its odds satisfy

$$\frac{p'_i}{p'_0} = \frac{q_i}{q_0} (1 + y\epsilon \tanh u_i), \quad (8.12)$$

so u_i is recovered by applying artanh to $\{p'_i q_0 / (p'_0 q_i) - 1\} / (y\epsilon)$. The image of the command interior has a smooth inverse. Boundary commands have zero probability under the uniform acquisition law.

Lemma 8.4 (Posterior Jacobian and unconditional density). *For $\epsilon > 0$,*

$$|\det D\Phi_{q,y}(u)| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \operatorname{sech}^2 u_i. \quad (8.13)$$

Under independent uniform commands and actual reports, the law of $p_{t,+}$ has a Lebesgue density bounded by $B_d \epsilon^{-d}$ at every $t \geq 1$, where one possible uniform constant is

$$B_d = 2^{-d} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1). \quad (8.14)$$

The bound is conditional on every preceding history as well as unconditional.

Proof. Write $r'_i = y\epsilon \operatorname{sech}^2 u_i$. In the d nonzero-state coordinates,

$$D\Phi_{q,y} = \operatorname{diag} \left(\frac{q_i r'_i}{D_y} \right) - \left(\frac{q_i r_i}{D_y} \right)_i \left(\frac{q_j r'_j}{D_y} \right)_j^\top.$$

The determinant lemma gives the determinant of the diagonal matrix times

$$1 - \sum_{i=1}^d \frac{q_i r_i}{D_y} = \frac{q_0}{D_y}.$$

Taking absolute values gives (8.13).

Before any conditioning on the sign, the joint density of the uniform command and that sign is $2^{-d} D_y / 2$. Change variables through the injective command map. Its contribution to the posterior density is

$$\frac{2^{-(d+1)} D_y^{d+2}}{q_0 \prod_{i=1}^d q_i \epsilon^d \prod_{i=1}^d \operatorname{sech}^2 u_i} \leq 2^{-(d+1)} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1) \epsilon^{-d}.$$

Sum the two signs, allowing their images to overlap, to obtain (8.14). Finally integrate the conditional bounds against the actual preceding-history law. The same bound survives because it is independent of that history. Thus no event probability or complementary command volume is discarded. $\square$

This is a full-density upper bound for the actual law, which is sufficient for a quantization lower bound. It is stronger than merely exhibiting a nonzero differential at one command, but it does not assert that every point of the invariant polytope is attained. Those are different statements.

8.4. A fixed codebook and the proof of the theorem. Project D_ϵ to its d coordinates p_+ . It lies in the cube $\nu_+ + [-a_\epsilon, a_\epsilon]^d$. Partition this cube into k^d cells, $k = \lfloor M^{1/d} \rfloor$, and select one belief in D_ϵ from each occupied cell. Fix those representatives once and for all, along with a first-minimum tie convention. Empty cells need no label. The discarded coordinate satisfies $p_0 = 1 - \sum_{i=1}^d p_i$. Hence within a cell the full-simplex ℓ^1 distance is at most twice the ℓ^1 distance of its d coordinates. The resulting codebook has at most M points and covering radius

$$r_M \leq \frac{4da_\epsilon}{k} \leq \frac{80d}{3}\epsilon M^{-1/d}. \quad (8.15)$$

Choose the nearest representative in full ℓ^1 distance. This preserves the covering bound and is Borel. The initial state is exactly ν ; thereafter the same codebook and the same update apply at every step.

Proof of Theorem 8.1. For $\epsilon > 0$, use (8.9) at the retained belief and then quantize with (8.15). Lemmas 8.2 and 8.3 justify the whole procedure on every history. By Corollary 7.2,

$$\sup_{t \geq 1} \|p_t - \hat{p}_t\|_1 \leq 3r_M \leq 80d\epsilon M^{-1/d}. \quad (8.16)$$

Equation (8.5) bounds the excess by $(\tau^2/d)(80d)^2\epsilon^2 M^{-2/d}$. This proves a pathwise upper bound for one machine, hence all the required mean and worst-history upper bounds. A checkpoint encoder can use the same final nearest-centre procedure.

For the lower bound, Lemma 8.4 and Lemma 6.2 give

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_j \|p_{t,+} - c_j\|_2^2 \geq \frac{1}{2}(2B_d v_d)^{-2/d} \epsilon^2 M^{-2/d}.$$

Multiply by τ^2/d and use Theorem 6.1. The bound holds for every $t \geq 1$ under the one fixed exploration law, with coding randomness allowed. Every causal encoder is a checkpoint encoder at each time, proving the causal lower bound for finite or infinite horizons. The inequality between mean and worst-history risks is immediate. When $\epsilon = 0$, both reports have probability $1/2$ independent of the state and command. Refreshing preserves the uniform preparation, so $p_t = \nu$ on every history. A single label predicts the terminal probe exactly relative to the full-history oracle. This proves the zero case and completes the uniform statement. $\square$

Corollary 8.5 (Persistent bits and double attenuation). *For b persistent bits, take $M = 2^b$ in (8.7). With the fixed terminal probe the squared excess has order $\epsilon^2 2^{-2b/d}$. If the actual terminal probe is instead*

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\epsilon \mathbf{1}_{\{i=j\}},$$

the order is $\epsilon^4 2^{-2b/d}$, with the same uniformity and the same filter. At $\epsilon = 0$, zero bits suffice in both models.

Proof. The first assertion substitutes the number of labels. In the second experiment the physical readout difference is multiplied by ϵ , while the acquisition process, posterior law, and filter are unchanged. Its squared excess is therefore multiplied by ϵ^2 . For the lower bound the same scaling applies to arbitrary decoder centres after clipping to the physical mean interval. The zero case was established above. $\square$

8.5. Feedback-uniform emulation in this model. For fixed command u , the probability of the positive report under belief p is

$$g_+(p, u) = \frac{1}{2} \left(1 + \epsilon \sum_{i=1}^d ((1 - \gamma)p_i + \gamma\nu_i) \tanh u_i \right).$$

For binary laws their total variation is the difference of their positive probabilities. Thus

$$\|g(\cdot \mid p, u) - g(\cdot \mid \tilde{p}, u)\|_{\text{TV}} \leq \frac{1}{2}\epsilon(1 - \gamma) \|p - \tilde{p}\|_1. \quad (8.17)$$

By Theorem 7.1, the actual report-path law and the finite-state reference report-path law obey

$$\|P_N^\alpha - \hat{P}_N^\alpha\|_{\text{TV}} \leq \min\{1, 40dN\epsilon^2(1 - \gamma)M^{-1/d}\} \quad (8.18)$$

for every common feedback controller. This statement is uniform over controllers but is finite-horizon; it does not identify infinite path laws. If a terminal probe is included in the compared observed path, its additional conditional error is at most $\tau \|p_t - \hat{p}_t\|_1$, averaged over its random index. This term is to be added to (8.18). Prediction regret for that probe instead uses its squared error, as in (8.5); the two claims are different.

The bound vanishes for the acquisition report law when $\gamma = 1$: each new state is independently uniform, so previous belief errors have no effect on the next acquisition report. The current posterior can still have nontrivial geometry and require finite prediction memory for the immediate terminal probe. This limiting case is a useful check on the meaning of each estimate.

8.6. Calibration and finite precision. The theorem is uniform over (8.1), but the codebook and update may use the known values of γ and ϵ . It is not a calibration-blind theorem. Numerically approximating each belief update by an admissible belief within η in ℓ^1 changes (8.16) to $3(r_M + \eta)$. The physical excess is at most $9\tau^2(r_M + \eta)^2/d$. Thus choosing η proportional to $\epsilon M^{-1/d}$ preserves the order. This is an accuracy requirement; it does not assert zero workspace or a universal bit-complexity bound for computing the update.

For rational calibration one can specify the invariant polytope by rational linear inequalities. Representatives can be selected in its occupied grid cells by a finite feasibility procedure; centres and a membership/tie rule form a finite read-only program. With nonrational calibration the theorem uses the declared calibrated real-number model; a numerical implementation must supply its own certified approximation. The only persistent acquired information after a report is its representative label.

Finally, the refresh lower bound and the fixed positive state floor are substantive. If γ tends to zero, the constant $2/3$ need not remain valid and the argument does not give a horizon-uniform theorem. That case is not hidden in the calibration interval.

9. NORMALIZED RESPONSE AND OBSERVATION MODULES

9.1. The covariance pairing. Let π be a probability measure and f, g bounded measurable functions. For u near zero define

$$\pi_u(dx) = \frac{e^{ug(x)}\pi(dx)}{\pi(e^{ug})}.$$

Proposition 9.1 (Normalized response). *The derivative of a normalized prediction is*

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \pi(fg) - \pi(f)\pi(g) = \text{Cov}_\pi(f, g). \quad (9.1)$$

More generally, if $L_u > 0$ is differentiable with the dominated derivatives needed for the quotient rule, then the derivative at u_0 of $\pi(fL_u)/\pi(L_u)$ is the covariance of f with $\partial_u \log L_u|_{u_0}$ under the normalized law at u_0 .

Proof. Differentiate numerator and denominator. At zero their values are $\pi(f)$ and one, and their derivatives are $\pi(fg)$ and $\pi(g)$. This proves (9.1). Apply the same quotient rule at u_0 and divide the derivative of L_u by L_{u_0} to obtain the general statement. Boundedness, or the stated domination in the general case, justifies integration of the derivatives. $\square$

For vector tests and several commands this gives a cross-covariance matrix. One index may represent an acquisition direction and the other a future test. That matrix is not automatically the Fisher information for an unknown physical parameter. In (8.9), command changes are selected by the observer; in Theorem 5.3, parameter changes vary the family of physical laws. The similar algebra does not equate the two statistical problems.

9.2. Finite positive functionals and exact product fibres. Suppose a real algebra of functions contains finite-dimensional spaces $U_1, \dots, U_L$, and let V contain all their products. A parameter is a linear functional $\ell \in \Omega \subset V^*$, where Ω is open in V^* . Positivity of measured blocks may define this open set. The observed bilinear forms are

$$B_\nu(f, g) = \ell(fg), \quad f, g \in U_\nu.$$

There are no cross-contact products unless the experiment actually measures them. Set

$$W = \sum_{\nu=1}^L \text{span}\{fg : f, g \in U_\nu\} \subset V.$$

Theorem 9.2 (Product-observation fibre). *The observation fibre through ℓ is*

$$(\ell + W^\perp) \cap \Omega. \quad (9.2)$$

Its local dimension is $\dim V - \dim W$. Recovery of every functional in Ω is equivalent to $W = V$. If the parameter domain is relatively open in an affine subspace with direction $L \subset V^$, the local invisible space is instead $L \cap W^\perp$.*

Proof. Two observations agree precisely when the difference functional vanishes on every generating product, equivalently on W . This gives (9.2). Openness allows all sufficiently small perturbations in the annihilator, so the local dimension is its dimension. The affine-domain version restricts those perturbations to L . $\square$

This is linear duality. The difficult model-specific work can lie in proving that these forms are actually observed, determining W , describing the locus where its dimension drops, and controlling real positive realizability. For the calibrated Hankel family of [16], V is a polynomial space and the observation is a normal compression of an inverse information matrix. The theorem supplies its abstract fibre mechanism, not the paper's complete degeneracy classification.

9.3. The contact calculation. For clarity, the local geometric interface can also be derived directly. Let $M \subset \mathbb{R}^k$ have an embedded smooth germ at x_0 , and let $R > 0$. Assume the local minimizing projection onto that germ computes the global distance in a neighborhood of x_0 . Define

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b).$$

Let Z have orthonormal columns spanning the Euclidean normal space at x_0 , and put $B = Z^\top R Z$.

Proposition 9.3 (Normal second jet). *Under these local projection assumptions,*

$$C_R(x_0 + h) = h^\top Z B^{-1} Z^\top h + o(\|h\|^2), \quad D^2 C_R(x_0) = 2 Z B^{-1} Z^\top.$$

Consequently the full second jet determines exactly the positive normal block B and has the tangent space as its kernel.

Proof. Replace the germ by its tangent space T to compute the quadratic term. The first-order minimizing condition for $\min_{v \in T} (h - v)^\top R^{-1} (h - v)$ says $R^{-1}(h - v) = Zn$ for some n . Hence $h - v = RZn$, and normal projection gives $Z^\top h = Bn$. The minimum is $(Z^\top h)^\top B^{-1} (Z^\top h)$. A smooth local parametrization of the germ has derivative spanning T ; its tangent Gram matrix in the positive R^{-1} metric is invertible. The implicit function theorem gives a differentiable minimizing parameter with the stated linear term. The quadratic expansion and the projection

assumption identify it with C_R near x_0 . Positive block inversion is one-to-one, proving the observation assertion. $\square$

If $R = JH(\theta)J^\top$ with Hankel entries $H(\theta)_{ab} = \theta_{a+b}$, then $v^\top R w$ evaluates the moment functional on the product of the two corresponding coefficient polynomials. Theorem 9.2 applies after this explicit identification. A higher-order distance jet has additional curvature and projection terms; replacing Sym^2 by Sym^m is not a derivation of a higher-order experiment.

9.4. Why general dynamic tests need not form a finite algebra.

Proposition 9.4 (Finite continuous algebras have finite quotients). *Let X be compact Hausdorff and let $W \subset C(X, \mathbb{R})$ be a finite-dimensional unital subalgebra. Then W is the algebra of all functions constant on a finite clopen partition of X . The number of nonempty cells is $\dim W$.*

Proof. For $f \in W$, sufficiently many powers of f are linearly dependent. A nonzero polynomial therefore vanishes on $f(X)$, so that image is finite. A finite basis of W has only finitely many joint values; its fibres form a finite clopen partition. Every member of W is constant on these fibres. Conversely, for a joint value a , choose for every distinct joint value b a basis function $f_{i(a,b)}$ separating them. The product

$$\prod_{b \neq a} \frac{f_{i(a,b)} - f_{i(a,b)}(b)}{f_{i(a,b)}(a) - f_{i(a,b)}(b)}$$

is in W and is the indicator of the fibre at a . These indicators are linearly independent and span all functions on the partition, which proves the result. $\square$

Thus a finite multiplication-closed continuous algebra is a useful model, but it cannot by itself represent arbitrary continuous predictive information. Growing test spaces and positive operator words are the more general primitive. Neither the algebra dimension nor one tangent rank replaces the actual predictive law or its quantization geometry.

10. EXPONENTIAL TRANSFORMS AND CONDITIONAL LIMIT INTERFACES

10.1. Random channels have nonlinear exponential pullbacks. For a Markov kernel $G : X \rightsquigarrow Y$, bounded measurable $f : Y \rightarrow \mathbb{R}$, and a fixed $\beta > 0$, define

$$\mathcal{T}_{G,\beta} f(x) = \beta^{-1} \log \int e^{\beta f(y)} G(dy | x). \quad (10.1)$$

For a probability P on X put $\mathcal{P}_{P,\beta}(h) = \beta^{-1} \log P(e^{\beta h})$. These expressions concern finite experiments and bounded sources; no long-time limit is presumed.

Theorem 10.1 (Exponential composition and entropy duality). *For $H : Y \rightsquigarrow Z$ and bounded f ,*

$$\mathcal{T}_{GH,\beta} f = \mathcal{T}_{G,\beta}(\mathcal{T}_{H,\beta} f), \quad (10.2)$$

$$\mathcal{P}_{PG,\beta}(f) = \mathcal{P}_{P,\beta}(\mathcal{T}_{G,\beta} f), \quad (10.3)$$

$$\mathcal{P}_{P,\beta}(h) = \sup_{Q \ll P} \{Qh - \beta^{-1} D(Q \| P)\}. \quad (10.4)$$

The supremum is attained at $dP^h/dP = e^{\beta h}/P(e^{\beta h})$. The source derivatives are

$$D\mathcal{P}_{P,\beta}(h)[u] = P^h u, \quad D^2\mathcal{P}_{P,\beta}(h)[u, v] = \beta \text{Cov}_{P^h}(u, v). \quad (10.5)$$

Proof. Exponentiate (10.2). Its right side becomes $G(H(e^{\beta f}))$, which is the definition of the composed kernel's exponential integral. Integrating the same identity against P proves (10.3). For any $Q \ll P$ of finite entropy,

$$D(Q \| P^h) = D(Q \| P) - \beta Qh + \log P(e^{\beta h}) \geq 0.$$

Rearrangement gives the upper inequality in (10.4), and $Q = P^h$ attains equality. Measures of infinite entropy do not improve the supremum for bounded h . Finally boundedness permits differentiation of the exponential integral in the directions u, v ; one quotient rule gives the mean and the second gives the covariance with the factor β . $\square$

If G is the deterministic map g , the exponential pullback is exactly $f \circ g$. If G is random, it is generally not Gf . Jensen gives $\mathcal{T}_{G,\beta}f \geq Gf$, with equality precisely when f is constant $G(\cdot | x)$ -almost surely at each relevant x . The entropy variational principle is classical [13]; the proof fixes its conventions for this experiment category rather than asserting a new variational principle.

Proposition 10.2 (Second-order chain rule for a random channel). *Put $h = \mathcal{T}_{G,\beta}f$ and*

$$G_f(dy | x) = \frac{e^{\beta f(y)} G(dy | x)}{G(e^{\beta f})(x)}.$$

Then the tilted joint law is $P^h(dx)G_f(dy | x)$ and

$$D\mathcal{T}_{G,\beta}(f)[u] = G_f u,$$

$$D^2\mathcal{T}_{G,\beta}(f)[u, v] = \beta \text{Cov}_{G_f}(u, v),$$

$$D^2\mathcal{P}_{PG,\beta}(f)[u, v] = \beta \text{Cov}_{P^h}(G_f u, G_f v) + \beta \mathbb{E}_{P^h} \text{Cov}_{G_f}(u, v).$$

The last expression is $\beta \text{Cov}_{P^h G_f}(u, v)$.

Proof. Multiplying the two tilted densities cancels $G(e^{\beta f})(x)$ and produces the original joint law multiplied by $e^{\beta f(y)}$ and its single total normalizer. Differentiation of the conditional normalizer gives the first two formulas just as in (10.5). Apply the ordinary second-order chain rule to (10.3). Its first term is the covariance of conditional means and its second is the mean of conditional covariances. The total covariance formula gives the final identity. $\square$

The conditional covariance term is essential. Discarding it would treat a random observation as a deterministic projection. This connects the positive-kernel and source-response layers exactly, while keeping them distinct from Fisher information for a model parameter.

10.2. A finite risk-sensitive recursion. Consider a known prepared predictive model with finite action sets, state update $F_t(s, a, y)$, report kernel $g_t(dy | s, a)$, bounded observed stage cost $c_t(s, a, y)$, and bounded terminal cost v_N . The cost here is an optimization objective, not an undeclared memory resource. For $\beta > 0$ define

$$v_t(s) = \min_{a \in A_t} \frac{1}{\beta} \log \int \exp\{\beta[c_t(s, a, y) + v_{t+1}(F_t(s, a, y))]\} g_t(dy | s, a). \quad (10.6)$$

Proposition 10.3 (Finite-horizon exponential dynamic programming). *The value (10.6) is the least value of $\beta^{-1} \log \mathbb{E} \exp(\beta[\sum_{j=t}^{N-1} c_j + v_N])$ over all causal strategies from s . A deterministic Borel minimizing policy exists with a fixed tie rule.*

Proof. Write $w_t = e^{\beta v_t} > 0$. Conditional expectation over the next report and then the future gives the multiplicative recursion $w_t(s) = \min_a \int e^{\beta c_t(s, a, y)} w_{t+1}(F_t(s, a, y)) g_t(dy | s, a)$. At the last step this is the exact conditional objective. Induct backwards; future randomized strategies cannot improve the already minimized continuation, and mixing current actions produces a convex combination of their positive objective values, no less than their minimum. Finite Borel action comparisons give a Borel first minimizer. Taking logarithms recovers (10.6). $\square$

This recursion needs a state sufficient for the future experiment and its cost. It cannot be applied to an arbitrary lossy statistic without proving closure. Compact infinite action sets would require appropriate existence and selection hypotheses; they are not needed for this proposition.

10.3. Deterministic LDP contractions. We record a conditional interface for later work. Suppose a sequence P_n on a Polish space has a good large-deviation principle with speed $a_n \rightarrow \infty$ and rate I . If T is continuous into another Polish space, its pushforward has rate

$$I_T(y) = \inf_{T(x)=y} I(x). \quad (10.7)$$

For an open or closed target set its inverse image has the same property, so the LDP lower and upper bounds give (10.7). Goodness follows because each finite-rate infimum is attained on compact sublevels and the sublevels of I_T are continuous images of those of I . This is the standard contraction principle [14]. It is not a consequence of the raw experiment axioms.

For a random channel one needs the joint law, a conditional rate, or a separate exponential approximation theorem. A small total-variation simulation defect does not supply these ingredients; an explicit example is given in Section 11.

10.4. Regular quadratic and labelled-phase interfaces.

Proposition 10.4 (Quadratic constrained reduction). *If $H > 0$ and A has full row rank, then*

$$\inf_{Ah=z} \frac{1}{2} h^\top H h = \frac{1}{2} z^\top (AH^{-1}A^\top)^{-1} z.$$

Proof. The Lagrange equations give $h = H^{-1}A^\top(AH^{-1}A^\top)^{-1}z$. The constraint matrix in parentheses is positive definite. Substitution gives the value, and strict convexity gives the unique minimizer. $\square$

Only if a separate probabilistic theorem identifies H^{-1} with the appropriate Gaussian covariance does this become a covariance contraction statement. Existence of an LDP alone is not that theorem.

For finitely many labelled components with common speed a_n , suppose component rates are I_j and their positive weights satisfy $-a_n^{-1} \log w_{n,j} \rightarrow c_j$, with $\min_j c_j = 0$. The joint labelled rate is $c_j + I_j$ and the unlabelled rate is $\min_j (c_j + I_j(x))$. To verify the bounds, bound a finite sum above by the number of summands times its largest summand and below by each summand; then apply the component LDPs. Compactness follows from the finite union of component compact sublevels. Dynamic prediction may still require the posterior over labels. A marginal log-sum is not automatically a Markov semigroup.

10.5. Closure and finite-dimensional memory. For a Markov kernel P and state map q , an exact reduced kernel $\bar{P}$ satisfies

$$P(f \circ q) = (\bar{P}f) \circ q.$$

Iterating this identity proves every finite-time identity, including the corresponding exponential transforms. If it fails, the projected state need not be Markov. For a finite-dimensional linear evolution split as

$$\dot{u} = A_{PP}u + A_{PQ}v, \quad \dot{v} = A_{QP}u + A_{QQ}v,$$

variation of constants in the second equation gives exactly

$$\begin{aligned} \dot{u}(t) &= A_{PP}u(t) + A_{PQ}e^{tA_{QQ}}v(0) \\ &\quad + \int_0^t A_{PQ}e^{(t-s)A_{QQ}}A_{QP}u(s) ds. \end{aligned} \quad (10.8)$$

The unresolved initial-state term is not optional. Unbounded generators, common form domains, and differentiable operator transports require separate analytic results; the finite-dimensional identity does not prove them. This is the precise boundary at which the history and memory program of [17] can be connected to the present foundations.

11. BOUNDARY EXAMPLES AND NON-IMPLICATIONS

The following examples specify the distinctions used in the theorems. They are part of the mathematical scope, not objections to studying stronger results under stronger hypotheses.

Example 11.1 (Less statistical information, more exact prediction labels). Let $Z \sim \text{Bernoulli}(1/2)$ and initially observe $X = Z$. The future target is again Z . Two labels retain the posterior mean exactly. Garble X into $Y = X + \xi$ with independent $\xi \sim N(0, 1)$. Bayes' formula and the ratio of the two Gaussian densities give

$$p(Y) = \mathbb{P}(Z = 1 \mid Y) = \frac{1}{1 + e^{1/2 - Y}}.$$

This is a smooth strictly increasing bijection from $\mathbb{R}$ to $(0, 1)$. The density of Y is positive everywhere, so the posterior law is continuous and has a positive density on every compact subinterval of $(0, 1)$. No finite set of centres can equal this posterior almost surely. More quantitatively, restrict to a compact interval where the density has a positive lower bound and apply Lemma 6.3 in one dimension. Every fixed M has strictly positive optimal excess squared error. Both oracles are different: absolute decision risk obeys statistical data processing, but excess coding risk relative to each oracle need not.

Example 11.2 (The same mean map with two noise geometries). Let $Z \sim N(\lambda, 1)$. For known $\epsilon > 0$, observing $Y_\epsilon = \epsilon Z$ is equivalent to observing Z by division, and its Fisher information for λ is one. Observing $\tilde{Y}_\epsilon = \epsilon\lambda + \xi$ with independent $\xi \sim N(0, 1)$ has Fisher information ϵ^2 . Both means equal $\epsilon\lambda$. Only the first covariance is scaled with the mean. Consequently a small singular value of a mean map does not determine its statistical information without the noise law.

Example 11.3 (Rank does not record global fibres or contact order). The derivative of $x \mapsto x^2$ is nonzero at 1 and -1 , yet their images coincide. For another distinction, let $Y_i = \lambda^r + \xi_i$, $\lambda \geq 0$, with independent unit Gaussian noise and $r \geq 2$. Every mean map has zero derivative at zero, but its local amplitude scale is $n^{-1/(2r)}$. Indeed the divergence between $\lambda = 0$ and a positive alternative is $n\lambda^{2r}/2$. Taking this bounded gives the lower resolution scale by two-point testing. For an upper bound use $\hat{\lambda} = (\bar{Y}_+)^{1/r}$ and $|x_+^{1/r} - b^{1/r}| \leq |x - b|^{1/r}$ for $b \geq 0$; its mean squared error is $O(n^{-1/r})$. The order of contact, not just the rank, matters.

Example 11.4 (Reachability is not attained mass). A protocol may permit all commands in an interval while the exploration law always chooses a single command. The set of legally attainable predictions can be continuous even if that particular exploration law visits only one prediction. Another exploration on the same command interface can spread positive mass over a full interval. Their average quantization risks differ. Theorem 8.1 avoids this ambiguity by specifying independent uniform commands and deriving their density.

Example 11.5 (Equal first two moments do not give equal laws). The two laws $\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$ and $\frac{1}{2}\delta_0 + \frac{1}{4}\delta_{-\sqrt{2}} + \frac{1}{4}\delta_{\sqrt{2}}$ have mean zero and variance one, but are different. Equality of a finite jet or covariance needs a model-specific identifiability theorem to imply anything stronger.

Example 11.6 (Fisher preservation need not be sufficiency). The Kagan–Shepp phenomenon, analyzed in [11], gives an insufficient statistic preserving Fisher information. We do not reproduce that external construction here or claim it as new. Its precise logical role is to prevent promotion of the equality case of Theorem 5.3 into the parameter-uniform recovery assertion of Theorem 5.2.

Example 11.7 (Small variation error can change the exponential rate). For $a_n \rightarrow \infty$ and $c > 0$, let

$$P_n = \delta_0, \quad Q_n = (1 - e^{-a_n c})\delta_0 + e^{-a_n c}\delta_1.$$

Their total variation is $e^{-an^c} \rightarrow 0$. At speed a_n , however, P_n gives rate $+\infty$ to 1, while Q_n gives it rate c . Thus even exponentially small variation error need not preserve a complete rate function. Appropriate superexponential approximation hypotheses are substantially stronger than an ordinary vanishing simulation defect.

Example 11.8 (Linear-source pressure can convexify a nonconvex rate). Let P_n put probability $1/2$ at each of -1 and 1 , for all n . At speed n its rate vanishes at those two points and is infinite elsewhere. Its limiting pressure for linear source sx is $|s|$. The Legendre transform of $|s|$ vanishes on the entire interval $[-1, 1]$. The full bounded-continuous-source Laplace functional and the restricted linear-source transform are not interchangeable.

Example 11.9 (Observation does not ensure Markov closure). Consider the deterministic chain $a \mapsto a, b \mapsto c, c \mapsto c$, with report $q(a) = q(b) = 0, q(c) = 1$. The current report 0 does not specify the next report: the two compatible hidden states give different answers. Thus no single reduced kernel works for every preparation. Future-test equivalence distinguishes a and b ; merely keeping their current report does not.

Example 11.10 (Zero evidence has no canonical normalized limit). On a two-point space the positive measures $(\delta, 0)$ and $(0, \delta)$ both tend to the zero measure as $\delta \downarrow 0$. Their normalized probabilities tend to different point masses. A continuous normalized extension at zero cannot be assigned from the zero measure alone.

Example 11.11 (A simulator may need delay memory). Let the unknown parameter be $\lambda \in \{0, 1\}$. A source with no control actions reveals λ at time one and constants thereafter. A target reveals a constant at time one and λ at time two. Their terminal information is equivalent. However, a parameter-independent simulator with only one retained state after time one, and no other retained tape, has the same internal state and the same new inputs under both parameters. Its second output therefore has the same law under both parameters and cannot equal the required two different point masses. Two states suffice by retaining the bit. If a target controller also retains its own state, the direct implementation is a product. Terminal sufficiency does not erase this state cost.

Example 11.12 (A finite command alphabet is a different experiment). At a fixed finite time, finite actions and finite reports produce only finitely many histories. With enough labels they can be retained exactly. The positive lower bound of Theorem 8.1 for all finite M uses the stipulated continuously distributed commands. Rounding the physical command alphabet itself changes that experiment. Approximating only the filter's arithmetic while observing the original experiment is instead the perturbation problem in Section 7.

These examples rule out four unqualified shortcuts: deriving global identifiability from local rank, deriving memory from information order, deriving statistical rates from a noiseless feature map, and deriving exponential asymptotics from ordinary variation approximation. The positive results in this paper specify the extra structure needed in each case.

12. INTERFACES WITH SINGULAR GEOMETRY AND DYNAMICS

The foundations established here do not require all experiments to be contractions of one physical path law or one rate function. They give a common language in which an actual map or causal simulator must be exhibited before a comparison is asserted.

12.1. What is established in this installment. The exact results are path construction, future-test minimality with stated realization conditions, charged simulation composition, posterior and entropy identities, the finite/dominated score identity, normalized response, product-observation fibres, checkpoint quantization, and exponential composition. The quantitative results are the

feedback- and stopping-compatible finite-state emulation theorem and the uniform dynamic memory law for the refreshing model. Their proofs occur in this article. The LDP and operator discussions are conditional interfaces, with a standard contraction principle and a finite-dimensional memory identity; no general Sinai or kinetic limit theorem is asserted here.

The principal dynamic model verifies a compact shrinking domain, actual probability mass, and stable updates directly. It gives one time-uniform isotropic degeneration law. It does not classify all anisotropic collisions, all hidden Markov filters, all priors, or weak-refreshing limits. The bounded comparison constants are not exact optimizer transition constants.

12.2. The role of the two geometric manuscripts. The finite-horizon manuscript [15] provides finer anisotropic profiles for its positive monomial experiments and a separate bounded observation-algebra law. Those hypotheses should remain attached to their conclusions. The present dynamic result is not a replacement for their collision geometry, and their fixed-horizon constants are not assumed uniform as the horizon grows.

The contact manuscript [16] provides a different geometric realization of information loss. Theorem 9.2 explains its linear-functional interface. A complete extension to singular statistical resolution still requires the actual noise metric, calibration, parameter nuisance, and admissible real domain. In particular independent measurement noise and correlated raw-data score noise can react differently to a small multiplication singular value. A complex determinantal component is not automatically a physically attainable real statistical submodel.

12.3. A precise next target. Let a parameter target be $\tau(\lambda)$ and let an actual feature map be $\mathbf{O}(\lambda)$. Its inverse modulus is

$$\omega(\delta) = \sup\{d(\tau(\lambda), \tau(\lambda')) : d_Z(\mathbf{O}(\lambda), \mathbf{O}(\lambda')) \leq \delta\}.$$

If a finite-memory procedure estimates that feature with a proved common error bound δ , fitting the model on a compact parameter set gives an error at most $\omega(2\delta)$: the true parameter is an admissible fit, and the triangle inequality bounds the discrepancy of the two model features. A matched statistical lower bound requires actual distributional closeness. If $\|P_{n,\lambda_0} - P_{n,\lambda_1}\|_{TV} \leq a < 1$, any estimator has maximum squared target risk at least

$$\frac{1-a}{8} d(\tau(\lambda_0), \tau(\lambda_1))^2.$$

Indeed choose the nearer of the two target points as a test; on an incorrect choice the estimation error is at least half their separation. The two error probabilities sum to at least $1 - a$.

These two elementary bounds specify a route to a resource–statistical singularity theorem, but do not prove a universal matched modulus law. Covering, testing, attainable alternatives, and tail control must be supplied. The geometry-of-rates literature, including [12], is the appropriate comparison, not a claim that all minimax theory follows from one rank locus.

12.4. The physical realization obligations remain. For deterministic hyperbolic and particle systems, one must still establish the true experiment kernels, physical clock handling, tightness and recovery at exponential scale, and the domains of any response or memory operators. A pathwise finite-state prediction error does not provide a Fourier local limit theorem. A static covariance factorization does not construct a collision process. A projected state does not acquire a Markov semigroup merely because it is named a predictive coordinate.

The historical typed-contraction program [17] can therefore be reorganized as a family of verified realization theorems attached to these interfaces. Its model-specific proofs are not assumed completed by the present definitions. In particular memory response in a varying Hilbert or form domain requires its own transport and domain arguments.

12.5. Conclusion. A common foundation is obtained by keeping the entire experiment ahead of its geometric representations. Positive preparation and future tests give prediction; a real acquisition law gives a distortion problem; stable updates give an online implementation; a causal simulator gives a comparison, with its state and error charged. In the refreshing model these steps yield a uniform finite-memory law through loss of the acquisition channel. The separate entropy, score, and exponential identities explain which forms of information transport are valid. Stronger singular and long-time results must enter through corresponding theorems, not through additional meanings assigned to the word “contraction”.

APPENDIX A. MEASURABILITY, FINITE COVERS, AND A SCALAR BENCHMARK

A.1. Finite-label implementation. A finite codebook $C = \{c_1, \dots, c_M\}$ in a metric space has a Borel nearest-centre selector whenever each distance $s \mapsto d(s, c_j)$ is Borel. Define its label by the least index attaining the minimum. The set on which label j is selected is the intersection of the finitely many sets $\{d(s, c_j) < d(s, c_i)\}$ for $i < j$ and $\{d(s, c_j) \leq d(s, c_i)\}$ for $i > j$. These sets are Borel. Thus the finite-label recursions used in this paper do not require an unproved measurable choice of a nearest point in an infinite set.

Codebook construction and codebook execution are different questions. Existence of finitely many representatives in nonempty cells is an ordinary finite choice. When an algorithmic construction is claimed, one must also supply a representation and a feasibility procedure for those cells. In Section 8, rational calibration gives a rational polytope; finite rational linear feasibility supplies this additional construction. This does not charge an unbounded list of history-dependent centres to a read-only program.

A.2. Conditional distributions and common versions. The existence of regular conditional distributions on standard Borel spaces is a classical input [4]. Its relevant conclusion is existence of a kernel under a fixed joint law. It does not provide one canonical version on every null history, nor does it alone give a version continuous in a parameter or uniformly valid on an uncountable family of null sets. The statements in the main text accordingly take one of three forms: an explicit positive finite-report formula; a compact continuous realization as in Theorem 3.3; or an already supplied jointly measurable predictive realization with its applicable quantifiers.

There is no hidden claim that all quotient images of a Borel space are standard Borel. The evaluation image into a countable product can be given its induced measurable structure, but to use ordinary kernel calculus on it one must verify that the image and descended maps have the required measurability. The compact theorem verifies precisely that fact in one useful class. Keeping the full posterior remains a valid alternative when compression of the state is not yet justified.

A.3. Quantization after a readout change.

Proposition A.1 (Bi-Lipschitz readout comparison). *Let S be a metric space with a probability ν , and let $p : S \rightarrow \mathbb{R}^q$ be an injective readout satisfying*

$$a d(s, t) \leq \|p(s) - p(t)\|_2 \leq b d(s, t), \quad a, b > 0.$$

If S has an M -point cover of radius r with centres in S , the checkpoint prediction risk is at most $b^2 r^2$. If some measurable map $T : \mathbb{R}^q \rightarrow \mathbb{R}^k$ has Lipschitz constant $L > 0$, and the law of $Tp(s)$ has density at most A , that risk is at least

$$\frac{1}{2L^2} (2Av_k M)^{-2/k}.$$

The second assertion does not require decoder centres to belong to $p(S)$.

Proof. For the upper bound map the cover centres through p and use the upper Lipschitz estimate. For any decoder centres $z_j \in \mathbb{R}^q$, the inequality $\|p(s) - z_j\|_2 \geq L^{-1} \|Tp(s) - Tz_j\|_2$ reduces the lower bound to Lemma 6.2. Take the infimum over the original centres. The constant a is useful for comparing intrinsic covers in both directions, but it is not needed in this particular density-based lower bound. In particular, the argument does not require inversion of a near-singular statistical covariance. $\square$

A linear projection or bounded coordinate map can be used for T . A coordinate change whose Lipschitz constant tends to infinity changes the bound; it cannot be called a harmless normalization at a degeneracy.

Proposition A.2 (Exact scalar uniform quantization). *For U uniform on an interval of length $a > 0$, the optimal squared M -centre quantization error is $a^2/(12M^2)$, for every integer $M \geq 1$.*

Proof. Order the centres. Their nearest-centre cells intersect the interval in intervals, allowing empty cells. If a cell has length l_j , its contribution is minimized by its midpoint and equals $l_j^3/(12a)$. Since $\sum_j l_j = a$, convexity gives $\sum_j l_j^3 \geq a^3/M^2$. Hence the total error is at least $a^2/(12M^2)$. Equal-length cells with their midpoints attain that value. Arbitrary randomized encoders cannot improve it by Theorem 6.1. $\square$

This exact formula is a benchmark for the meaning of label cardinality, not a claim that the dynamic posterior density in Section 8 is uniform. The latter is bounded by its actual Jacobian and can depend on the preceding history. Its two-sided order, rather than an exact optimizer constant, is what we prove.

A.4. Predictable error budgets and random horizons. The stopped version of Lemma 4.2 is often more useful than the deterministic sum. In a comparison with common visible stopping rule σ , append an absorbing terminal symbol. The two conditional kernels coincide once the symbol has appeared. Their discrepancy at time t is therefore bounded by $\mathbf{1}_{\{t < \sigma\}} e_t(H_t)$, a measurable function of the common prefix. The telescoping proof integrates it against the actual prefix law. This gives the expectation of the active-step sum, without replacing a random horizon by an uncharged successful count.

If the active-step sum is bounded by a deterministic budget B on every path, its expectation is at most B . If only an expected budget is known, that is the exact kind of budget furnished by the theorem. A tail-budget statement requires an additional probability estimate. These are distinct resource certificates even though the same positive summands occur in all three.

APPENDIX B. EXPLICIT FINITE CHECKS AND THE LIMITS OF COMPUTATION

B.1. A two-state posterior calculation. The smallest dynamic instance has $d = 1$, $\gamma = 4/5$, and $\epsilon = 1/4$. Put $v = \tanh u$ and select $v = 1/2$, which corresponds to the legal command $u = \operatorname{artanh}(1/2) < 1$. Starting from the uniform preparation, the pre-report belief is $q = (1/2, 1/2)$, and

$$\mathbb{P}(Y = +1) = \frac{17}{32}, \quad \mathbb{P}(Y = -1) = \frac{15}{32}.$$

The posterior probabilities of state 1 are respectively

$$p_1^+ = \frac{9}{17}, \quad p_1^- = \frac{7}{15}.$$

Thus the same current command produces two different posteriors. Discarding the negative report or assigning its conditional law mass one would alter the acquisition law used in the proof. With the fixed terminal probe the corresponding success probabilities are

$$\frac{1}{2} + \frac{1}{4} \frac{9}{17} = \frac{43}{68}, \quad \frac{1}{2} + \frac{1}{4} \frac{7}{15} = \frac{37}{60}.$$

The mean posterior agrees with the pre-report probability:

$$\frac{17}{32} \frac{9}{17} + \frac{15}{32} \frac{7}{15} = \frac{1}{2}.$$

This is the posterior martingale property for the report step; the refresh step has its own affine prediction and is not being omitted.

The derivative with respect to v , rather than u , at this point for the positive report is

$$\frac{q_0 q_1 \epsilon}{D_+^2} = \frac{16}{289}.$$

Multiplying by $dv/du = 1 - v^2 = 3/4$ gives $12/289$, exactly (8.13) in one dimension. These distinct derivatives are recorded because replacing a physical command by a nonlinear coordinate without its Jacobian would change the preparation density.

B.2. A direct resource calculation. Suppose a report converter has two persistent states (for example, it retains the previous binary report), the target predictor has M labels, and a feedback controller has D labels. The implemented composite uses at most $2MD$ triples. On the next step it reads the old triple, the current input, and fresh independent randomness, and writes the next triple. The source history may be longer, but it is not consulted by this implementation. Any reduction below $2MD$ needs a proved state identification; it is not obtained by calling the converter a sufficient statistic.

Conversely, if a converter is memoryless and deterministic, its state set has one element. In that special case it need not enlarge the target label count. This is the correct context for a nonincrease assertion. It does not imply the false general statement about exact predictive memory refuted in Example 11.1.

B.3. A finite precision requirement rather than a free real tape. For $d = 1$ the invariant domain is an interval inside the probability simplex. A codebook can be constructed using interval intersections, and its labels are ordinary finite integers. One evaluates the two rational Bayes fractions, with the command likelihood evaluated to a specified accuracy, then selects the nearest admissible representative. The perturbation form of Theorem 7.1 accounts for the incurred state error. Since denominators are at least $3/4$, no division by a vanishing report probability occurs.

For general fixed d , the same procedure uses a finite polytope feasibility construction. It may be computationally expensive. Our cardinality theorem does not state an optimal running time or workspace bound. The program may store its predetermined codebook, as permitted by the read-only model; it may not put acquired commands or reports in that program. The current command must be discarded after updating the label, and the terminal decoder may not reread it as free side information.

B.4. Role of the accompanying diagnostics. The source package includes exact-rational checks of Bayes normalization, the odds inverse, the rank-one determinant identity, the refresh contraction inequality at a finite family of inputs, and the scalar quantization identity. It also checks a finite conditional-covariance identity, a feedback path-law comparison, and specified deliberately incorrect formulas. These computations are reproducibility aids for signs, constants, and implemented examples.

The universal statements in this article are proved analytically above. A finite test does not establish measurability, a global differential bound, a minimax lower bound, or uniformity over all times. In particular, no number of successful finite diagnostics certifies that every possible hidden Markov model has the displayed memory law. Build receipts, source hashes, and diagnostic output belong to the accompanying delivery record rather than to the logical assumptions of any theorem.

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